

**“Studies on chemical mediated induction of
resistance in chickpea against dry root rot
(*Rhizoctonia bataticola*).”**

THESIS

Submitted to the

**Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya,
Gwalior**

**In Partial Fulfillment of the Requirements
For the Degree of**

MASTER OF SCIENCE

In

AGRICULTURE

(PLANT PATHOLOGY)

By

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2013

CERTIFICATE I

This is to certify that the thesis entitled “**Studies on chemical mediated induction of resistance in chickpea against dry root rot (*Rhizoctonia bataticola*)**” submitted in partial fulfillment of the requirements for the degree of **MASTER OF SCIENCE IN AGRICULTURE (PLANT PATHOLOGY)** of Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior is a record of the bonafide research work carried out by **Mr. Amrit Kumar Pipalde** under my guidance and supervision. The subject of the thesis has been approved by the Student's Advisory Committee and Director of Instruction.

No part of the thesis has been submitted for any other degree or diploma (Certificate awarded etc.) or has been published/published part has been fully acknowledged. All the assistance and help received during the course of the investigation has been acknowledged by him.

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CERTIFICATE II

This is to certify that the thesis entitled “**Chemical mediated induction of resistance in chickpea against dry root rot (*Rhizoctonia bataticola*).**” submitted by **Mr. Amrit Kumar Pipalde**, I.D. No. A/IN/85/2006 to Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior, in partial fulfillment of the requirements for the degree of **Master of Science in Agriculture** in the Department of **Plant Pathology**, has been, after evaluation, approved by the External Examiner and by the Student’s Advisory Committee after an oral examination on the same.

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ACKNOWLEDGEMENT

I praise God for enabling me to accomplish this great task of thesis work in his grace for his glory. I find no words to express my sincere feelings of gratitude towards my most esteemed guide and chairperson of my advisory committee, **Dr. R.K. Singh** Assistant Professor, Department of Plant Pathology, College of Agriculture, Indore, who deserves my most sincere thanks and respect for inspiring and excellent guidance, contact, encouragement, unceasing interest, constructive criticism and helping attitude throughout the investigation and preparation of the manuscript.

I extend my sincere gratitude to **Dr. Ashok Krishna** Professor and Head of section Department of Plant Pathology and other member of my advisory committee **Dr. S.P. Mishra**, Associate Professor, Department of Plant Pathology and **Dr. R.K. Choudhary**, Associate Professor, Department of Entomology, College of Agriculture, Indore, for their suggestions and helping attitude during the research work.

With profound respect, I wish to express my sincere gratitude to **Dr. A. M. Rajput** Dean, College of Agriculture, Indore, (M.P) for providing necessary facilities during the experiment.

I also thank Shri Shankar Choudhary, Lab technician, Department of Plant Pathology, Shri Radheshyam ji fard, pradeep chouhan for extending the help during my research work.

I thank to all my seniors Vishal sir, Firoz Khan, Rajesh Belwanshi, Rajesh Patel, Kamlesh, Rajender, Nitesh, Deepak, Sandeep, Rakesh, Akriti, R. Garg, and my friends R. sitole, M. Gehlot A. Choursia, A. Parmar, M.Parmar, Rajiv, Gajendra, Asharam and my juniors R. Parmar, Pal, Shyam, Tanvi, Dharmendra and Ajay and others, who helped me directly or indirectly during course of investigation.

I find no rhetorical gems from the ocean of words to express my profounding feeling to most venerable parents Shri Bhadar Pipalde and Smt. Kadvi Bai and my grandfather (late) Ratanlal Pipalde, who were always present in all of my good and the bad times. I also want to express my profounding feelings to my sister Radha Pipalde and my brother Sunil Pipalde who have piloted me up to this stage and whose love, devotion, blessing and care throughout my life enable me to achieve this seemingly invincible goal.

Date:

Place: Indore

(AMRIT KUMAR PIPALDE)

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LIST OF ABBREVIATIONS

Legends	=	Description
<i>et al.</i>	=	And others
%	=	Percentage
/	=	Per
SA	=	Salicylic acid
°C	=	Degree celcius
cm	=	Centimeter(s)
m	=	Meter
m ²	=	Square meter
mm	=	Milli meter
INT	=	Isonicotinamide
Fig	=	Figure
Kg	=	Kilogram(s)
cv	=	Cultivar
PDA	=	Potato dextrose agar
viz.,	=	Wide list
i.e.	=	In reference to, that is
Sem±	=	Standard error of means
CD	=	Critical difference
ANOVA	=	Analysis of variance
S.S	=	Sum of squares
D.F	=	Degree of freedom
S.V	=	Source of variation
M.S.S	=	Mean sum of square
FS/FA	=	Foliar spray/ Foliar application
DAS	=	Day after sowing
Q/ha	=	Quintal/hectare
PPM	=	Parts per million
SAR	=	Systemic acquired resistance

CHAPTER- 1

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is one of the most important pulse crops grown in the semi arid and tropical climate. It originated from middle East and now grown in 45 countries across the world. There are two main types as desi (small in size having brown colour) and kabuli (bold seeded with light brown colour). It is a rich source of protein, vitamins and minerals containing 17-22% protein, 60-64% carbohydrate and 3-4% fat (Sindhu *et al.*, 1974) alongwith a good source of phosphorus and calcium. Chickpea leaves contain malic acid which bear medicinal value in intestinal disorders.

India grows chickpea on about 8.56 million ha area producing 7.35 million tonnes, which represents 38% and 50% of national pulse acreage and production, respectively. Chickpea production has gone up from 3.65 to 7.35 million tonnes between 1950-51 and 2009-10. During the period, area has also increased from 7.57 to 8.56 million ha and the productivity has steadily increased to 858 kg/ha from 482 kg/ha. Notwithstanding its distribution throughout the country, six states viz., Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Karnataka and Andhra Pradesh together contribute 92% of the production and 95% of the area of the country.

Among several constraints responsible for low productivity, wilt caused by *Fusarium oxysporum* f. sp. *ciceris* and dry root rot caused by *Rhizoctonia bataticola* are the destructive diseases. Use of bio-agents undoubtedly is an eco-friendly and sustainable approach for management of wilt and root rot complex.

Major losses in chickpea yield are attributed mainly due to soil borne pathogens. The crop suffers from a number of soil borne pathogens like dry root rot *Rhizoctonia bataticola*, collar rot *Sclerotium rolfsii* and wilt – *Fusarium oxysporum* f.sp. *ciceris*. Among these wilt caused by *Fusarium oxysporum* Schl. Emend, Snyd and Hans f. sp. *Ciceris* (Padwick) Snyd and Hans is common and important throughout the world (Gupta *et al.*, 1987). Dry root rot infected by *Rhizoctonia bataticola* (Taub), Butler, is another serious disease at early crop stage.

In recent years, induced resistance following treatment with biotic and abiotic factors has been considered to be a potential approach for the control of plant diseases. Therefore, induced resistance holds promise as a new technology for the control of plant diseases and has been proven to be effective in the laboratory and in a few field cases. Resistance inducing chemicals that are able to induce broad spectrum of disease resistance, offer an additional option for the farmers to complement with genetic disease resistance.

Induced resistance is defined as an enhancement of the plants defensive capacity against a broad spectrum of pathogens and pests that is acquired after appropriate stimulation. Pre-treatment of plants with a virulent pathogens (biotic inducers) or chemical compounds (abiotic inducers) can enhance resistance to subsequent attack not only at the site of treatment, but also in tissues distant from the initial infection sites. Typically, this inducible resistance system known as systemic acquired resistance (SAR) is effective against diverse pathogens including viruses, bacteria and fungi (Ryals *et al.*, 1996).

Dry root rot of chickpea has been reported from India, Iran, Australia, Ethiopia, Pakistan, Spain and USA. The importance of this disease in Mediterranean region is not known.

Drying of the plants appear suddenly in the field at flowering and podding. There is no drooping which is commonly seen in the plants affected by wilt. The leaves and stems of affected plants are straw-colored. The roots are usually dry, unless the soil is wet. The tap root is quite brittle, show shredding of the bark and can be broken easily. Minute dark black sclerotial bodies can be seen with the aid of a 10 X hand lens on the surface of the root, as well as in the pith.

The disease appears suddenly when ambient temperatures are between 25 and 30 °C.

Pathogen: *Rhizoctonia bataticola* (Taub) Butler=*Macrophomina phaseolina* (Maubl.) Ashby.

The fungus lacks fruiting bodies and spores. The mycelium is light-brown, thick in which black sclerotia are formed. Sclerotia are variable in form, small and loosely connected by mycelial threads.

The fungus is soil-borne and may survive in the soil in the form of sclerotia for long time. The fungus has a wide host range and can infect the roots of several crops. Therefore, crop rotation is not effective to control the disease.

Since the disease is soil-borne the development of sick plot is essential for resistance screening. However, the presence of other soil-borne pathogens in sick plot cannot be avoided. To confirm resistance specifically to *R. bataticola* it would be necessary to follow laboratory technique.

Various chemicals have been discovered that seem to act at various points, these defense activating networks mimic all or part of the biological activation of resistance. In the present investigation an attempt was made to correlate the appearance of wilt in different treatment of chemical SAR compounds with following objectives:

Objectives:

- Evaluation of SAR inducing chemicals against *Rhizoctonia bataticola* (*in vitro*).
- Evaluation of SAR inducing chemicals for control of dry root rot in pot experiment.
- Evaluation of SAR inducing chemicals for control of dry root rot in the field at different concentrations.

CHAPTER- 2

REVIEW OF LITERATURE

Studies on the isolates of *Rhizoctonia bataticola* were undertaken on the collection of the diseased specimens, isolation and identification of the pathogen and compatibility of two SAR compounds salicylic acid and isonicotinamide with *R. bataticola*, to impart resistance for the pathogen. Information on these aspects is being reviewed and presented in this chapter.

Brief resume of work done in M.P., India and abroad:

2.1 General:

Kodandaramaiah (1983) reported that soaking of grains in/or spraying of plant with one of two vitamins (nicotinamide or ascorbic acid) was generally associated with marked decrease in the activity of protease enzyme concurrently with increasing the protein level indicating that vitamins could alleviate the inhibitory effects of salt stress by enhancing protein synthesis where vitamins might act as activators for protein synthesis.

Metraux *et al.* (1991) reported that INA was effective in inducing systemic resistance in cucurbits. It was also effective in minimising fungal and bacterial diseases of foliage in pear, pepper, rice and tobacco under field conditions.

Delaney *et al.* (1995) reported that salicylic acid is a natural phenolic compound present in many plants, and an important component in the signal transduction pathway involved in local and systemic resistance to pathogens.

Lawton *et al.* (1995) reported that debate continues about the role of salicylic acid in the process of induced systemic resistance.

Vernooij *et al.* (1995) reported that exogenous application of SA was shown to be mimicking certain aspects of a pathogen infection, resulting in both SAR gene expression and induction of SAR.

Dann and Devera (1996) proved that 2,6-dichloro-isonicotinic acid (INA) effective in decreasing foliar diseases in green beans *Phaseolus vulgaris*.

Colson-Hanks *et al.* (2000) reported that 2, 6 -dichloroisonicotinic acid or benzothiadiazole reduced the incidence of root infection also increased fresh and dry weight of shoots and roots.

Matta (2001) reported that SAR (systemic acquired resistance) is a type of induced resistance that follows necrogenous infections and is characterized by accumulation of salicylic acid and increased synthesis of pathogenesis related proteins.

Kubo and Sato (2002) proved that salicylic acid (SA), a chemical inducer of SAR, and SAR gene expression were induced by high-temperature treatments in greenhouse-grown cucumbers, providing evidence for the existence of SAR to heat stress.

Baehler *et al.* (2005) showed that the microbial and plant phenolic metabolite salicylate was found to interfere with the expression of both 2, 4-diacetylphloroglucinol (DAPG) and pyoluteorin (PLT). The developed reporter fusions will be a highly valuable tool to study *in situ* expression of this bacterial biocontrol trait on plant roots, i.e. at the site of pathogen suppression.

El-Bassiony (2005) reported that presoaking of seeds with optimal concentration of vitamins has been shown to be beneficial in seedling growth under saline condition by increasing physiological availability of water and nutrient.

Choudhary *et al.* (2007) showed that several specific strains of species, i.e. *B. amyloliquifaciens*, *B. subtilis*, *B. pasteurii*, *B. cereus*, *B. pumilus*, *B. mycoides* and *B. sphaericus* elicit significant reduction in the incidence or severity of various diseases on a diversity of hosts.

Molinari (2007) reported that new developments in understanding the role of SA in plant defence are reported from studies on root response to soil pests, such as root-knot nematodes.

Bassuony (2008) showed that application of nicotinamide or ascorbic acid mitigates the adverse effects of salinity through increasing the synthesis of protein and decreasing protease activity and corrects the nutritional disorders induced by salinity by decreasing Na⁺ ions and increasing K⁺, Ca²⁺ and Mg²⁺ ions contents over those of control plants and salinized ones. Sodium ions were accumulated while K⁺, Ca²⁺ and Mg²⁺ ions were significantly decreased in salt stressed *Zea mays* plant.

Kremer (2009) suggested the use of biological products such as those containing the plant defense compound salicylic acid and amino acids by foliar application to improve resistance to root pathogens.

Dhore *et al.* (2011) reported that condensation of substituted 1,3,4-oxadiazoles and isoniazide gives N-((3,5-substituted)-4H-1,2,4-triazole-4yl) isonicotinamide(D1-4) for the IR, ¹H-NMR, and nitrogen analysis characterized all these compounds. The newly synthesized compounds were screened for antibacterial and antifungal activity.

Doornbos *et al.* (2011) elucidated how much the bacterial rhizosphere microflora of *Arabidopsis* is affected by induced systemic resistance (ISR) or systemic acquired resistance (SAR). Therefore, the bacterial microflora of wild-type plants and plants affected in their defense signaling was compared. Additionally, ISR was induced by application of methyl jasmonate and SAR by treatment with salicylic acid or benzothiadiazole. As a comparative model, we also used wild type and ethylene-insensitive tobacco. Some of the *Arabidopsis* genotypes affected in defense signaling showed altered numbers of culturable bacteria in their rhizospheres; however, effects were dependent on soil type. Effects of plant genotype on rhizosphere bacterial community structure could not be related to plant defense because chemical activation of ISR or SAR had no significant effects on density and structure of the rhizosphere bacterial community.

Patel and Hemantaranjan, (2012) inferred that the significant response of applied SA at the rate of 1.5 mM under drought condition was relatively more in ICC 4958 genotype in terms of antioxidant response; substantially alleviated drought induced adverse effect and yield. It is also concluded that

drought stress at 50% flowering was found to be more damaging than the 50% podding phase.

2.2 SAR+Pathogen:

Greenberg *et al.* (2000) proposed that *acd5* plants mimic *P. syringae*-infected wild-type plants and that both SA and ethylene are normally involved in regulating cell death during some susceptible pathogen infections.

Roetschi *et al.* (2001) reported that *pad2*, a mutant with reduced production of the phytoalexin camalexin, was hypersusceptible. The accumulation of salicylic acid (SA) and PR1 protein was strongly reduced in *pad2*.

El-Ganaieny (2002) studied that effect of antioxidants aminobutyric acid (ABA), potassium salicylate (PS), oxalic acid (OA), salicylic acid (SA) and ascorbic acid (AA) *in vitro* on fungi *Fusarium oxysporum*, *F. solani* and *Fusarium moniliforme* [*Gibberella fujikuroi*]. for growth and spore germination tests, 0, 2, 4, 6, 8 and 10 mM concentration of ABA, PS, OA, SA or AA. Spore germination was greatly reduced by many tested chemicals at 8 mM concentration or less. The inhibitory effect of the antioxidants increased with increasing concentrations.

Yu Ting Chen, *et al.* (2007) showed that the combined treatment of pear fruit with *Cryptococcus laurentii* with salicylic acid SA at 100 µg/ml resulted in a remarkably improved control of *Penicillium expansum* and *Botrytis cinerea* infections, including the pre-inoculated *Penicillium expansum*, in comparison with the application of *Cryptococcus laurentii* or SA alone.

Ali *et al.* (2009) observed the efficacy of some plant resistance elicitors vis: chitosan (CHI), salicylic acid (SA) and hydroquinone (HQ) in comparing to the fungicide Rhizolex T-50 as seed treatments showed significant reduction in the fungal growth *in vitro*.

Amal (2009) evaluated the inhibitory effect of twelve chemicals from different groups of known resistance inducers to the growth of bean seed-borne fungi was evaluated, *in vitro*, at concentrations of 1, 5, 10 and 20 mM. the fungal growth was negatively affected by increasing concentrations of the

tested inducer chemicals. benzoic, salicylic and sorbic acids completely suppressed fungal growth at 10 mM.

Abdel-Monaim and Ismail (2010) evaluated the four antioxidant compounds (coumaric acid, citric acid, propylgalate and salicylic acid each at 100 and 200 ppm) for their *in vitro* and *in vivo* effect against *Fusarium* pathogenic isolates caused root rot and wilt diseases in pepper plants. All chemicals reduced spore formation in both *Fusarium spp.* at 100 or 200 ppm and the inhibitory effect of antioxidants increased with increasing their concentrations.

Angela Mika *et al.* (2010) showed that four enzymes pmPOX3, pmPOX2a, pmPOX2b and pmPOX1 were differentially but strongly stimulated by methyl jasmonate, salicylic acid, and elicitors (*Fusarium graminearum* and *Fusarium culmorum* extracts, and chitosan) indicating their function in pathogen defence.

Gayatrivedi *et al.*, (2012) showed that two isoforms of catalases in shoot extract (CAT-IS and CAT-IIS) and single isoform in root extract (CAT-R) were detected in chickpea. CAT-IS and CAT-R activities were inhibited in respective extracts treated with pathogen whereas, CAT-IIS activity was not inhibited.

2.3 SAR+Disease:

Seguchi *et al.* (1992) described that N cyanomethyl-2-chloroisonicotinamide as a related compound with an apparently similar action in controlling rice blast disease.

Gaffney *et al.* (1993) reported that seed treatment with salicylic acid and Bion induces systemic resistance in chickpea seedlings against *Fusarium* wilt.

Kogel *et al.* (1994) reported that isonicotinic acid decreased powdery mildew infections in barley.

Hijwegen and Verhaar (1995) reported that isonicotinic acid decreased powdery mildew infections in cucumber.

Hijwegen *et al.*(1996) reported that INA decreased powdery mildew development in roses.

Shakirova *et al.* (2000) reported that salicylic acid (SA) was an effective growth regulator, improving the resistance of wheat to fungal diseases, activating growth, and increasing yield.

Wang ChangChun *et al.* (2003) revealed that SA and ethylene enhanced resistance of tobacco to black shank and attenuated the spread of the pathogen. Treatments of continuous watering plant for 4 days or soaking leaf for 24 h with KCN did not restore the resistance lost in nahG transgenic plant. This result supported the hypothesis that KCN was not able to restore the resistance to fungal diseases lost in nahG transgenic plant.

Ishikawa *et al.* (2005) reported that plants sprayed with validamycin A (VMA) or validoxylamine A (VAA) accumulated salicylic acid and had elevated expression of the systemic acquired resistance (SAR) marker genes P4 (PR-1), Tag (PR-2), and NP24 (PR-5). Foliar spray of VMA also controlled late blight and powdery mildew of tomato.

Pandey and Kalra (2005) examined that selected groups of resistance activator viz. Isonicotinamide, 2-chloronicotinic acid, 5-nitrosalicylic acid, 4-chlorosalicylic acid, DL-2 aminobutyric acid, 2-aminobutyric acid, O-acetylsalicylic acid, 4-amino salicylic acid, salicylic acid as soil drench using 3 week old seedlings transplanted to root-knot nematode infested pots. Maximum reduction in root-knot severity and nematode population occurred with 4-chlorosalicylic acid, O-acetyl salicylic acid, 2-chloronicotinic acid and gave significant flower yield advantages.

Asghari *et al.* (2007) showed that hot solution of SA effectively controls fruit decay organisms and reduces fruit storage disorders. Fruit dipped only in hot water of 45 °C without SA treatment were severely attacked by decay organisms and suffered from storage disorders especially of storage scald and softening.

Tian ShiPing *et al.* (2007) reported that mainly introduce the controlling effect of exogenous SA treatment on postharvest diseases in different fruits and discussed the physiological response of fruits to SA treatment. These results are important for fully understanding the function of SA in the defence response of fruits to pathogen infection.

Raju *et al.* (2008) revealed that the seeds of susceptible cultivar L-550 pretreated with salicylic acid (SA), spermine (Spm) and SA+Spm delay the onset of disease symptoms, whereas the seeds treated with water showed complete collapse within 25 days of treatment.

Amal (2009) observed that salicylic acid significantly reduced the disease incidence by 73.33 and 50.00% in case of *F. solani* f. sp. *phaseoli* and *M. phaseolina*. *Sclerotium rolfsii* were less frequent.

Morsy (2011) showed that salicylic acid, K₂HPO₄ and neem oil respectively gave a significant $P \leq 0.05$ protection against rust, downy mildew, wilt and root rot diseases in both tested cvs. Ismailia-1 and wade-1 when applied as plant spraying and/or seed soaking compared with check treatment, whereas, Bio-Zaid and Bio-Arc were less effective.

2.4 SAR+Biogent:

Pieterse *et al.* (1996) reported that systemic resistance in *Arabidopsis* induced by bacteria was claimed to be independent of salicylic acid accumulation and pathogenesis-related gene expression.

Saikia *et al.* (2003) reported that selected isolates of *Pseudomonas fluorescens* (Pf1-94, Pf4-92, Pf12-94, Pf151-94 and Pf179-94) and chemical resistance inducers (salicylic acid, acetylsalicylic acid, DL-norvaline, indole-3-carbinol, and lichenan) were examined for growth promotion and induced systemic resistance against Fusarium wilt of chickpea (cv. JG-62). A significant increase in shoot and root length was observed in *P. fluorescens* treated plants. The isolates of *P. fluorescens* systemically induced resistance against Fusarium wilt of chickpea caused by *F. oxysporum* f.sp. *ciceri*, and significantly ($P=0.05$) reduced the wilt disease by 26-50% compared to the control.

Singh *et al.* (2003) showed that treatment of chickpea seeds with *Pseudomonas* spp. increased the synthesis of cinnamic, ferulic and chlorogenic acids, which showed antifungal effects against *Sclerotium rolfsii*.

Gupta *et al.* (2008) showed that under *in vitro* tests, all the plant activators up to 2000 ppm concentration exhibited their compatibility with the PGPR strains tested. Upon assaying of elicitors with plant-pathosystem, disease suppression was significantly ($P=0.05$) high with integrated

application of PGPR strains and plant activators as compared to their individual applications.

Saman Abeysinghe (2009) reported that *Pseudomonas aeruginosa* 7NSK2 and SA44, soil application of salicylic acid, previously reported ISR inducers in bean, also significantly reduced disease development in greenhouse pot experiments.

2.5 SAR+Fungicide:

El-Mougy, (2004) reported that salicylic acid (SA) and acetylsalicylic acid (ASA) in addition to Rizolex-T, seed dressing or soil drench at the rate of 2, and 3 g/kg, respectively, have significantly reduced the percentage of root rot incidence at both pre-, and post-emergence stages of lupin plants growth comparing with untreated control in artificially infested soil with *Fusarium solani*, *Rhizoctonia solani* and *Sclerotium rolfsii*. Lupin plants treated with SA and ASA showed more activity in chitinase than Rizolex-T treatment.

Ali *et al.* (2009) reported that efficacy of some plant resistance elicitors viz: chitosan (CHI), salicylic acid (SA) and hydroquinone (HQ) in comparing to the fungicide Rhizolex T-50 as seed treatments showed significant reduction in the fungal growth *in vitro*.

2.6 SAR+ Cultivar:

Bayraktar and Dolar (2002) investigated the role of exogenous salicylic acid (SA) application on inducing resistance to *A. rabiei* in chickpeas was investigated. SA was applied to seeds and foliage of the susceptible chickpea cultivar, Cantez-87. The *in vitro* effect of SA on mycelial growth, spore germination and germ tube growth of *A. rabiei* was also determined. With the application of SA to seeds, infection by *A. rabiei* was reduced at a rate of 5.23-23.44% compared to the controls. The highest inhibition was observed in plants with 1 mM SA application. One application of SA to foliage 2 days before the inoculation with the pathogen decreased the infection more compared to applications on other days; at 0.8 mM, the disease severity was inhibited by 48.15%. The inhibition rate was 46.10% with the twice 5 mM SA application. In almost all the cases, the most effective dose was found out to be 5 mM. SA also was applied following inoculation with the pathogen and the disease severity reduction values at one day after inoculation ranged

from 2.55 to 43.37%. The most effective dose was 2.5 mM. Fungitoxicity studies revealed that application of 7.5 mM SA under in vitro conditions inhibited the colony growth, spore germination and germ tube growth.

Zahid *et al.* (2005) reported that the seeds of chickpea variety AUG-424, susceptible to *Fusarium oxysporum* f. sp. *ciceri* (FoC), were surface sterilized with sodium hypochlorite, thoroughly rinsed with sterilized water and immersed in two concentrations of salicylic acid (1.0 & 1.5 mM) and Bion (0.3 & 0.4 mM). On the basis of disease rating done after root cutting, wilt incidence was significantly less in chemically treated plants as compared to control ones. Fresh and dry weights of shoot and root were higher in treated plants as compared to control especially in plants grown from Bion treated seeds.

Coram and Pang, (2007) Used microarray technology and a set of chickpea (*Cicer arietinum* L.) unigenes and grasspea (*Lathyrus sativus* L.) expressed sequence tags, chickpea responses to treatments with the defence signalling compounds salicylic acid (SA), methyl jasmonate (MeJA) and aminocyclopropane carboxylic acid (ACC) were studied in three chickpea genotypes with ranging levels of resistance to ascochyta blight [*Ascochyta rabiei* (Pass.) L.]. The *A. rabiei* resistant chickpea genotypes showed a more substantial range of defence-related gene induction by all treatments, indicating that they may possess stronger abilities to resist pathogens. Further, the involvement of SA, MeJA and ACC signalling was identified for the regulation of some important *A. rabiei* responsive transcripts, as well as cross-talk between these pathways. suggest the involvement of *A. rabiei*-specific signalling mechanisms for the induction of several transcripts that were previously implicated in *A. rabiei* resistance.

Raju *et al.* (2007) observed that the induction of systemic acquired resistance (SAR) by salicylic acid (SA), spermine (Spm) and SA + Spm was studied in the chickpea cultivars L-550 and ICCV10, which were susceptible and resistant to wilt disease caused by *Fusarium oxysporum* f.sp. *ciceri* [*Fusarium oxysporum* f.sp. *ciceris*], respectively. The levels of H₂O₂ were higher in roots and shoots of the susceptible cultivar L-550 than that of the resistant cultivar (ICCV-10). The activities of catalase (CAT), peroxidase

(POD), ascorbate peroxidase (APX) and superoxide dismutase (SOD) were enhanced significantly after treatment with SA, Spm and SA + Spm in both cultivars. The inductions of multiple isoforms of CAT were observed in roots and shoots of the susceptible cultivar and they were insensitive to SA treatment.

Raju *et al.* (2008) investigated the induction of some defense related enzymes and phenolics in roots and shoots of two different genotypes of chickpea which were susceptible (L 550) and resistant (ICCV 10) to wilt disease treated with salicylic acid, spermine (Spm), SA+Spm and *Fusarium oxysporum* f. sp. *ciceri* was investigated. Higher levels of polyphenol oxidase (PPO), phenylalanine ammonia-lyase (PAL), 1, -3-glucanase (PR-2) and phenolics were observed in roots and shoots of resistant cultivar than that of susceptible cultivar on treatment with elicitors and the pathogen.

Raju. *et al.* (2009). elicitation of proteases and protease inhibitors (PIs) by salicylic acid (SA) and spermine (Spm) was investigated in roots and shoots of two different genotypes of chickpea cultivars ICCV10 and L550, which were resistant and susceptible to wilt disease, respectively. SA and Spm were found to suppress the elicitation of proteases in the resistant cv, whereas they induce it in susceptible cv. Elicitation of new trypsin and chymotrypsin inhibitors was observed in the roots and shoots of resistant cv treated with SA and Spm. Antifungal property of root extract of resistant cv increased following treatment of seedling with SA and Spm compared with susceptible cv.

Zahid *et al* (2010) studied induced systemic resistance in chickpea against wilt disease caused by *Fusarium oxysorum* f. sp. *ciceri* (FOC) was studied by treating the seeds with benzo (1,2,3)-thiadizole-7-carbothioic acid-s-methyl ester (Bion), salicylic acid (SA) and di-potassium hydrogen phosphate (K₂HPO₄). Reduction in disease was observed in both type of applications but seed dressing was found more effective than soaking method. Highest reduction, 63% in wilt disease was observed with Bion dressing followed by SA, 40% and K₂HPO₄, 30%. Bion and SA showed 41 & 24% reduction in the disease, respectively.

Jayalakshmi *et al.* (2011) observed that phenylalanine ammonia-lyase activities were increased by two fold in resistant cv. A1 than that of susceptible cv. JG62 upon treatment with salicylic acid and *T. harzianum* L1. Induction of the multiple isoforms of catalases was observed in roots and shoots of cv. JG62 and they were insensitive to salicylic acid (SA) or *T. harzianum* L1.

Zahid *et al.* (2011) grew chickpea variety, C727 susceptible to *Ascochyta rabiei*, was grown in small plots of 20 sq feet area. Induction treatments were given by spraying aqueous solutions of 1.0 mM Salicylic acid, 50 mM K₂HPO₄ (analytical and commercial grade), 0.4 mM Bion and two types of neem leaves extract (50 mg/ml, leaves boiled in methanol, methanol was evaporated then residue dissolved in water and in other extract neem leaves were boiled in water) at flowering stage of the plants in triplicates. Control plants were treated with water only. One week after induction treatments, all the plants were challenged with spore suspension of *A. rabiei* (106 spores/ml). Blight disease was significantly reduced in chickpea variety C727 after spraying the plants with all the chemicals tested. Maximum reduction in the disease was obtained with Salicylic acid, followed by Bion. Slight increase in yield was observed in Bion, K₂HPO₄ (commercial) and neem extract treated plants as compared to control plants but the difference was non-significant. A significant increase in total proteins and peroxidase activity was observed after induction treatments with all the chemicals tested.

Ghazanfar *et al.* (2011) investigated the role of resistance-inducing substances (chemicals and plant extracts) in three chickpea cvs. C-44, Pb-91, and Bittle-98 in field experiments against *Ascochyta* blight disease. These cultivars were selected on the basis of better yield potential shown in previous experiments. Aqueous solution of salicylic acid (SA) at 0.5, 1.0, and 1.5 mM; acibenzolar-S-methyl (BionReg., ASM) at 0.4, 0.8, and 1.2 mM; and KOH at 25, 50, and 75 mM were applied, whereas *Azadirachta indica* A. Juss., *Datura metel* L., and *Allium sativum* L. plant extracts were applied at 5, 10, and 15%. Disease reduction data was recorded at different intervals from 4 to 14 d after being induced and inoculated with the pathogen. Overall

results revealed that significant disease reduction (79.3%) was provided by ASM in the cv. C-44 at 1.2 mM compared with SA, whereas the least significant was KOH. Maximum disease reduction (43.5%) against disease in the plant extracts was observed by applying neem leaf extract, but *D. metel* and *A. sativum* extracts were not effective.

Chapter 3

Material and methods

The present studies on “chemicals mediated induction of resistance in chickpea against dry root rot (*Rhizoctonia bataticola*).” As SAR/ISR compounds and response of these chemicals to different inoculum load and substratum were carried out in the laboratory, under pot condition and in field experiments in the Department of Plant Pathology, College of Agriculture Indore, M.P. The details of materials used, experimental procedure followed and techniques adopted are given here as under:

3.1 Materials

The materials used included seeds of chickpea variety (JG 62), chemicals for culture media, SAR chemicals, glass wares, equipments and few miscellaneous articles.

3.1.1 Glasswares

Standard "Borosil" make glasswares like Petri dishes, beakers, funnels, pipettes, Erlenmeyer flasks, sterilized cotton, culture tubes, measuring cylinder etc. were used during the course of study.

3.1.2 Equipments and miscellaneous articles

Equipments used during course of investigation included research Binocular microscope, refrigerator, autoclave, hot air oven, BOD incubator, Laminar air flow, weighing balance, LPG gas burner and hot plate. Small instruments like inoculation needle, scalper, razor, glass cavity slides, glass rod, cork borer, wash bottle, forceps, scissor, knife, polypropylene bags, desiccators, glass marking pencils, cover slips, brush, dropper, match box, plastic pots etc.

3.1.3 Chemicals

(a) Cleaning solution

Cleaning solution contained potassium dichromate 80 g, distilled water 300 ml and concentrated sulphuric acid 400 ml. This was used to clean the glass wares.

(b) Mercuric chloride solution

A 1:1000 solution of HgCl_2 was prepared and used for surface sterilization of samples.

(c) Mounting medium

Lacto phenol and cotton blue with the following composition were used as the mounting media for studying the characteristics of the pathogen.

(i) Lacto phenol

Phenol (pure crystals, liquefied to 20 ml by gentle heating on a water bath)

Lactic acid	20 ml
Glycerol	40 ml
Distilled water	20 ml

(ii) Cotton blue

Anhydrous lacto phenol	67 ml
Distilled water	20 ml
Cotton blue	0.1 g

3.1.4. Media

Ingredients of the culture media used in the study have been given below:

Potato dextrose agar medium

Potato dextrose agar (PDA) medium having the following ingredients was used for a few physiological studies (Dhingra and Sicclair, 1993).

Peeled and sliced potato (extract)	200 g
Dextrose	20 g
Agar -agar	18 g
Distilled water	1000 ml
pH (adjusted to)	6.5

Hi- Media made PDA medium was used during all the physiological studies.

3.2 Evaluation of SAR chemicals for control dry root rot in the field:

3.2.1 Field studies

A trial to evaluate the effect of two SAR chemicals (salicylic acid and isonicotinamide) against the control of *R. bataticola* infection under natural condition in disease prone area was conducted.

3.2.2 Experimental details

The present experiment consisting of 9 treatment combinations was carried out in field. Details of the experiment are given below.

a) Design : Randomized Block Design

b) Treatments : 9

T₁: Seed treatment with 200ppm salicylic acid.

T₂: Seed treatment with 400ppm salicylic acid.

T₃: Seed treatment with 200ppm isonicotinamide.

T₄: Seed treatment with 400ppm isonicotinamide.

T₅: Seed treatment with 200ppm salicylic acid and foliar spray of salicylic acid with 200ppm at 30 DAS.

T₆: Seed treatment with 400ppm salicylic acid and foliar spray of salicylic acid with 400ppm at 30 DAS.

T₇: Seed treatment with 200ppm isonicotinamide and foliar spray of isonicotinamide with 200ppm at 30 DAS.

T₈: Seed treatment with 400ppm isonicotinamide and foliar spray of isonicotinamide with 400ppm at 30 DAS.

T₉: control (without any seed treatment and foliar spray).

c) Replications : 3

d) Total plot number : 9x3=27

e) Size of plot : 2.0x1.8 = 3.6m²

f) Variety : JG-62

g) Distance (P-P) : 30cm x 30cm

(R-R) : 10cm x 10cm

3.2.3 Observations recorded

1. Total number of germinated plants.
2. Total number of diseased plant at 30, 60 & 90 DAS.

3.2.4 The diseased specimens

Chickpea plants exhibiting symptoms of dry root rot from the field and pots were collected at seedling and adult stages, brought to the laboratory and symptoms on collar and root tissues of plants exhibiting typical drying and rotting with discoloration were examined by naked eyes. Periodic isolation of the pathogen was also made and identification was done on the basis of characters described in monograph on *Rhizoctonia bataticola*. (Ashby.,1927).

The per cent disease incidence was calculated by using the following formula.

$$\text{Per cent disease incidence} = \frac{\text{Number of plants affected}}{\text{Total number of plants observed}} \times 100$$

3.2.5 Test organism/ Isolation and purification of the pathogen

The pathogen *Rhizoctonia bataticola* was isolated from root region of the diseased plants by time segment method (Rangaswamy, 1972) and purified by hyphal tip method and maintained on potato dextrose agar stants. The affected portions (roots) of the diseased plants were cut with the help of sharp razor and rinsed with sterilised water to remove traces of dirt. These were surface sterilised by dipping in 1:1000 mercuric chloride solution for one minute and washed twice with sterile water. These pieces were transferred aseptically to sterilised Petri dishes containing solidified PDA-s in a laminar air flow. The Petri dishes were incubated at 26±1°C. The appearing fungus was observed after 72 hours and isolations were made from developing colonies for further study.

3.2.6 *Rhizoctonia bataticola*

The morphological, cultural and formation of sclerotia were the principle characters to identify the pure cultures of *R.bataticola*. The characters were compared with those described by Ashby (1927) and the fungus was identified as *R.bataticola*.

3.2.7 Multiplication of fungal inoculums

The fungal inoculum was multiplied on sterilized sand maize meal medium in 1000 ml flasks containing 400 g in each flask. A loop full of fungus from actively growing culture was inoculated into each flask and incubated at room temperature for 15 days. The fungal culture multiplied on sand maize meal medium was used for inoculation to the soil.

3.3 Methods

3.3.1 Cleaning and sterilization of glassware

The glass wares were cleaned by dipping them in cleaning solution for 5 minutes and finally rinsed with running tap water for 30 minutes. The Petri dishes were sterilized in a hot air oven at $180 \pm 2^{\circ}\text{C}$ for 1.5 to 2 hours. The inoculation needle, cork borer and other metallic instruments were sterilized by dipping them in alcohol and heating red hot over flame of the spirit lamp/Bunsen burner.

3.3.2 Preparation of media

(i) Potato dextrose agar

Peeled potatoes cut into triangular pieces were boiled in $\frac{1}{2}$ liter water in a sauce pan for 8-10 minutes. In another pan, agar-agar was boiled in $\frac{1}{2}$ liter water separately so that it gets completely dissolved. Potato extract was collected after filtration on muslin cloth over a funnel and added to dissolved agar-agar along with dextrose. The medium was poured in conical flask of 250 ml, plugged with non absorbent cotton and autoclaved at 121.6°C (15 lbs) for 25 minutes. Streptomycin sulphate @ 1.0 g/l was added to the medium before pouring into Petri dishes to avoid bacterial contamination. It was used during the investigation.

3.4.1 In vitro testing of SAR chemicals against *Rhizoctonia bataticola*. (Salicylic acid and Isonicotinamide) by following poison food technique

Two SAR/ISR compounds namely salicylic acid and isonicotinamide along with a control were screened against *Rhizoctonia bataticola in vitro* by poisoned food technique (Nene and Thapliyal, 1993).

The required quantities (200ppm, 400ppm, 600ppm and 1000ppm) of the SAR/ISR compounds were added separately in Erlenmeyer flasks containing molting sterilized PDA and cooled to 55°C so as to give the desired concentration and poured in sterilized Petri plates. Three replications were maintained for each treatment. After solidification of medium each plate was inoculated by transferring a 6 mm disc of 7 days old culture of the *Rhizoctonia bataticola* containing spores and mycelial grown on PDA. Suitable controls without the addition of fungicide were also maintained. The plates were incubated at 28 + 2°C. The data with respect to the radial growth and mycelial characters were recorded at the 7 days after inoculation or full expansion of colony growth in control.

3.5.1 Evaluation of SAR chemicals in pot experiment (Pots filled with soil prior inoculation with *Rhizoctonia bataticola*)

A pot culture experiment was carried out for the chemicals mediated induction of resistant in chickpea against dry root rot (*Rhizoctonia bataticola*) in the poly house at College of Agriculture, Indore. Chickpea seeds were sowed in the earthen pots containing sick soil, the commercially available SAR chemicals like salicylic acid and isonicotinamide were used in the experiment. Sterilized pot mixture was taken in six kg capacity pots and chickpea seeds were sown separately @ 20 seeds per pot. Before sowing of seeds, commercially available SAR chemicals viz., salicylic acid and isonicotinamide applied to the seed treatment and foliar spray at their respective dosages as well as in combinations into the respective sets of pots. Simultaneously *Rhizoctonia bataticola* were inoculated of sorghum multiplied 15 g culture per kg of soil, in the pots. The treatments were replicated four times in completely randomized design. A suitable control was also maintained. Observations on diseased plant at post emergence stage were taken after 05, 12, 20 and 27 DAS as follows.

3.5.2 Treatment details:-

Treatments : 9

T₁: Seed treatment with 200ppm salicylic acid.

T₂: Seed treatment with 400 ppm salicylic acid.

- T₃: Seed treatment with 200ppm isonicotinamide.
- T₄: Seed treatment with 400ppm isonicotinamide.
- T₅: Seed treatment with 200ppm salicylic acid and foliar spray of salicylic acid with 200ppm just after germination of the plants.
- T₆: Seed treatment with 400ppm salicylic acid and foliar spray of salicylic acid with 400ppm just after germination of the plants.
- T₇: Seed treatment with 200ppm isonicotinamide and foliar spray of isonicotinamide with 200ppm just after germination of the plants.
- T₈: Seed treatment with 400ppm isonicotinamide and foliar spray of isonicotinamide with 400ppm just after germination of the plants.
- T₉: control (without any seed treatment and foliar spray).

Design : CRD

Replications : 4

Total number of pots : 108 (3 pots per replication)

3.6 Statistical analysis

The procedure for analysis for Randomized Block Design (RBD) and CRD (for *in vitro* experiments) were employed to analyses the data. The critical differences were worked out at 5% probability level. The values were transformed by $x + 0.5$ (Panse and Sukhatme,1978).

The experiments were conducted in the research field and PG laboratory, Department of Plant Pathology, College of Agriculture, Indore (M.P.).

Chapter 4

Result

Results of the research work on “Investigations on chemical mediated induction of resistance in chickpea against dry root rot (*Rhizoctonia bataticola*).” carried out using variety JG-62 in the Department of Plant Pathology, College of Agriculture, Indore, during Rabi 2011-12 have been presented in the following paragraphs.

4.1 Symptoms of the disease

Drying of the plants appear suddenly in the field at flowering and podding stage. There is no drooping which is commonly seen in the plants affected by wilt. The leaves and stems of affected plants are straw-colored. The roots are usually dry, unless the soil is wet. The tap root is quite brittle, show shredding of the bark and can be broken easily. Minute dark black sclerotial bodies can be seen with the aid of a 10 X hand lens on the surface of the root, as well as in the pith. The disease appears suddenly when ambient temperatures are between 25 and 30 °C.

Pathogen: *Rhizoctonia bataticola* (Taub) Butler=*Macrophomina phaseolina* (Maubl.) Ashby

4.2 Isolation of the causal organism

The pathogen *Rhizoctonia bataticola* was isolated from root region of the diseased plants by time segment method (Rangaswamy, 1972) and purified by hyphal tip method and maintained on potato dextrose agar slants.

4.3.1 *Rhizoctonia bataticola*

The morphological and cultural characteristics and formation of sclerotia were the principle observations recorded to identify the pure cultures of *R.bataticola*. The characters were compared with those described by Ashby(1927) and the fungus was identified as *R.bataticola*.

The fungus lacks fruiting bodies and spores. The mycelium is light-brown, thick, in which black sclerotia are formed. Sclerotia are variable in form, small and loosely connected by mycelial threads.

4.4.2 Multiplication of fungal inoculums

The fungal inoculum was multiplied on sterilized sand maize meal medium in 1000 ml flasks containing 400 g of the medium in each flask. A loop full of fungus from actively growing culture was inoculated into each flask and incubated at room temperature for 15 days. The fungal culture multiplied on sand maize meal medium was used for inoculation of the soil.

4.5 Incidences of dry root rot in chickpea under the influence of chemical elicitors

Two chemical elicitors i.e. salicylic acid and isonicotinamide were applied on chickpea variety JG-62 at 200ppm and 400ppm as seed treatment alone and in combination with their foliar application at 30 DAS in a field experiment to see their effect on imparting resistance against dry root rot (as a reaction in the incidence of the disease).

The data on incidence of dry root rot (%) 30, 60 and 90 DAS have been presented in table 4.1 the data on disease incidence 30 DAS among different treatments showed significant differences.

i. Incidence of dry root rot 30DAS

The minimum disease incidence (8.25) was observed in application of seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS which was statistically identical to the incidence in seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (9.99). The next higher disease incidence (11.60) was observed in seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS followed by seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (11.75) which were statistically at par and also with the disease incidence (14.07) in seed treatment with 200ppm isonicotinamide.

Fig:-1 Incidence of dry root rot in chickpea under the influence of chemical elicitors.

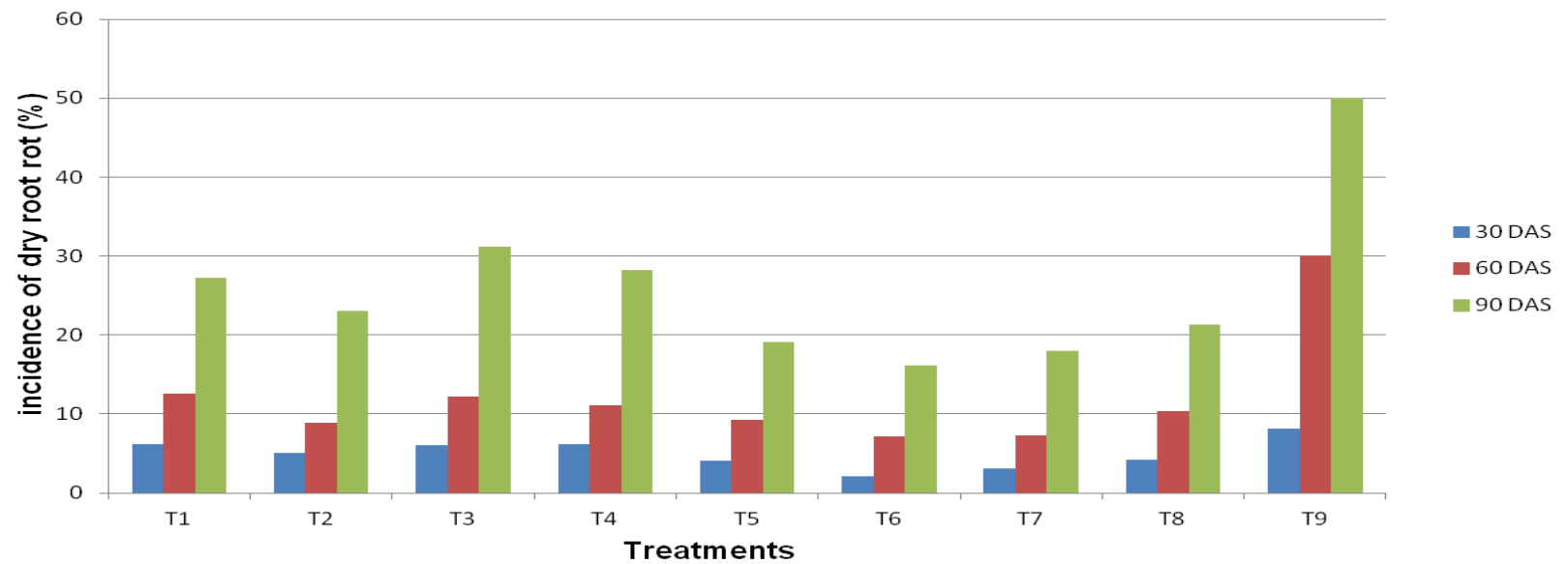


Table 1: Incidence of dry root rot in chickpea under the influence of chemical elicitors

Treatments	Treatment Description	Mean incidence of dry root rot (%)		
		30 DAS	60 DAS	90 DAS
T1	ST with 200ppm SA	6.15 (14.32)*	12.56 (20.74)	27.20 (31.44)
T2	ST with 400ppm SA	5.12 (13.07)	8.89 (17.29)	23.05 (28.66)
T3	ST with 200ppm INT	6.02 (14.07)	12.27 (20.47)	31.15 (33.89)
T4	ST with 400ppm INT	6.12 (14.30)	11.09 (19.44)	28.17 (32.03)
T5	ST with 200ppm SA and FS of SA with 200ppm at 30 DAS	4.07 (11.60)	9.27 (17.72)	19.08 (25.88)
T6	ST with 400ppm SA and FS of SA with 400ppm at 30 DAS	2.07 (8.25)	7.15 (15.37)	16.17 (23.44)
T7	ST with 200ppm INT and FS of INT with 200ppm at 30 DAS	3.05 (9.99)	7.31 (15.68)	18.01 (25.11)
T8	ST with 400ppm INT and FS of INT with 400ppm at 30 DAS	4.16 (11.75)	10.31 (18.67)	21.28 (27.40)
T9	Control	8.14 (16.55)	30.03 (33.21)	50.01 (45.01)
	Sem±	0.81	0.91	1.52
	CD	2.43	2.74	4.57

* The data transformed by Angular transformation.

A disease incidence of (14.32) was recorded in seed treatment with 200ppm salicylic acid and seed treatment with 400ppm isonicotinamide (14.30) which was statistically at par with the earlier set of treatments i.e. seed treatment with 400ppm salicylic acid (14.07) and seed treatment with 200ppm isonicotinamide (13.07).

The maximum disease incidence (16.55) was recorded in control which was statistically higher than all the treatments except seed treatment with 200ppm isonicotinamide (13.07), seed treatment with 200ppm salicylic acid (14.32) and seed treatment with 400ppm isonicotinamide (14.30).

Thus salicylic acid at 400ppm applied as seed treatment along with its foliar application at 30 DAS resulted in minimum incidence of dry root rot and thus has shown slight resistance against the disease.

The next effective treatment was salicylic acid at 200ppm applied as seed treatment coupled with foliar application at 30 DAS.

ii. Incidence of dry root rot 60 DAS

The minimum disease incidence (15.37) was observed in application of seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS which was statistically identical to the incidence in seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (15.68), seed treatment with 200ppm salicylic acid (17.29) and seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (17.72). The next higher disease incidence (17.29) was observed in seed treatment with 200ppm salicylic acid followed by seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (17.72) seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (18.67) and seed treatment with 400ppm isonicotinamide (19.44). Still higher incidence (18.67) was observed in seed treatment with 200ppm salicylic acid which was statistically non-significant with the disease incidence (19.44) in seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS, (20.47) seed treatment with 200ppm isonicotinamide and (20.74) seed treatment with

200ppm salicylic acid respectively. A disease incidence of (18.67) was recorded in seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS which was statistically at par to seed treatment with 400ppm isonicotinamide (19.44), seed treatment with 200ppm isonicotinamide (20.47), seed treatment with 200ppm salicylic acid (20.74), respectively.

The maximum disease incidence (33.21) was recorded in control which was statistically higher than all the treatment.

Thus salicylic acid at 400ppm applied as seed treatment along with its foliar application at 30 DAS resulted in minimum incidence of dry root rot and thus has shown slight resistance against the disease.

The next effective treatment was salicylic acid at 200ppm applied as seed treatment coupled with foliar application at 30 DAS.

iii. Incidence of dry root rot 90 DAS

The minimum disease incidence (23.44) was recorded in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS which was statistically identical to the incidence in seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (25.11), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (25.88), and seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (27.40). The next higher disease incidence was recorded in seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (25.11) followed by seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (25.88), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (27.40) and seed treatment with 400ppm salicylic acid (28.66) which were statistically at par. The next higher disease incidence (27.40) was recorded in seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS followed by seed treatment with 400ppm salicylic acid (28.66) and seed treatment with 200ppm

salicylic acid (32.44) which were statistically at par. Still the next higher disease incidence (28.66) was recorded in seed treatment with 400ppm salicylic acid followed by seed treatment with 200ppm salicylic acid (31.44) and seed treatment with 400ppm isonicotinamide (32.03). A disease incidence of (33.89) was observed in seed treatment with 200ppm isonicotinamide which was statistically at par with the earlier set of seed treatment with 200ppm salicylic acid (31.44) and seed treatment with 400ppm isonicotinamide (32.03). The maximum disease incidence was (45.01) was observed in control which was statistically higher than all the treatments. The minimum disease control by control (45.01).

Thus salicylic acid at 400ppm applied as seed treatment along with its foliar application at 30 DAS resulted in minimum incidence of dry root rot and thus has shown slight resistance against the disease.

The next effective treatment was salicylic acid at 200ppm applied as seed treatment coupled with foliar application at 30 DAS.

The maximum disease incidence (16.55), (33.21) and (45.01%) and were recorded in control at 30, 60 and 90 DAS, respectively.

Thus salicylic acid at 400ppm applied as seed treatment along with its foliar application at 30 DAS (8.25), (15.37) and (23.44) recorded at 30 DAS, 60 DAS and 90 DAS, respectively resulted in minimum incidence of dry root rot and thus has shown slight resistance against the disease.

The next effective treatment was seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS. (9.99), (15.68) and (25.11) recorded at 30 DAS, 60 DAS and 90 DAS, respectively.

4.6 influence of chemical elicitors on germination percent of chickpea at 10 DAS

Two chemical elicitors i.e. salicylic acid and isonicotinamide were applied on chickpea variety JG-62 at 200ppm and 400ppm as seed treatment alone in a field experiment to see their effect on inhacing germination (%) of chickpea (Plate No. 1).

The data of germinated chickpea plants (percent) as recorded at 10 DAS was subjected to statistical analysis. The analysis of variance computed from these data is given in table 2. Two chemical elicitors i.e. salicylic acid and isonicotinamide exhibited that their different concentrations did not produce any significant difference in the germination (%) of chickpea at 10 DAS.

The study of the data revealed that the minimum germination percent (43.62) was found in control which was statistically identical to seed treatment with 200ppm isonicotinamide (45.17), seed treatment with 200ppm salicylic acid (47.93), seed treatment with 400ppm salicylic acid (51.71), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (51.95), seed treatment with 400ppm isonicotinamide (52.30) and seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (53.35). The maximum germination percent (55.60) was found in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS which was statistically at par with the earlier set of seed treatment with 200ppm salicylic acid (47.93), seed treatment with 400ppm salicylic acid (51.71), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (51.95), seed treatment with 400ppm isonicotinamide (52.30) seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (53.35) and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (55.43).

Thus the data revealed that there were statistically non-significant difference between all the treatments but the highest germination was achieved by the seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS followed by seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS.



Plate 1:- Field view of experimental site

Fig:-2 influence of chemical elicitors on germination percent of chickpea at 10 DAS.

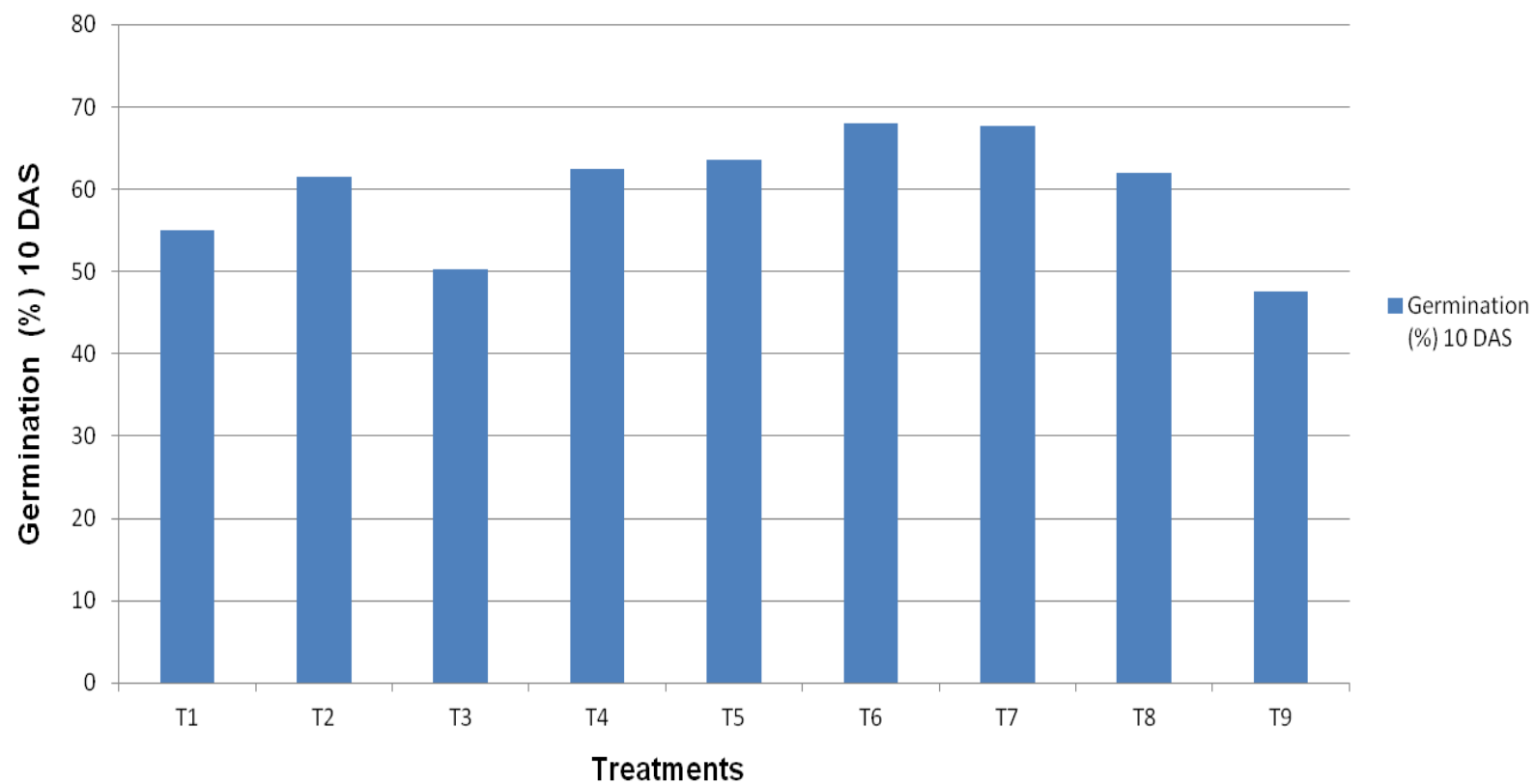


Table 2: influence of chemical elicitors on germination percent of chickpea at 10 DAS

Treatments	Treatment Description	Germination (%) 10 DAS
T1	ST with 200ppm SA	55.09 (47.93) *
T2	ST with 400ppm SA	61.52 (51.71)
T3	ST with 200ppm INT	50.29 (45.17)
T4	ST with 400ppm INT	62.46 (52.30)
T5	ST with 200ppm SA and FS of SA with 200ppm at 30 DAS	63.51 (53.35)
T6	ST with 400ppm SA and FS of SA with 400ppm at 30 DAS	67.95 (55.60)
T7	ST with 200ppm INT and FS of INT with 200ppm at 30 DAS	67.72 (55.43)
T8	ST with 400ppm INT and FS of INT with 400ppm at 30 DAS	61.99 (51.95)
T9	Control	47.60 (43.62)
	Sem±	3.29
	CD	9.86

*The data transformed by Angular transformation.

4.7 Radial growth of *Rhizoctonia bataticola* as influenced by chemical elicitors

For studying the reaction of 2 chemical elicitors against *R. bataticola* by poisoned food technique was followed (Nene and Thapliyal, 1993). Four concentrations of chemical elicitors were used exhibiting an increasing trend of colony diameter when the concentration of the chemical elicitor decreased than the normal dose.

The data on colony diameter was recorded at 7 days of incubation. The fungus grew in its characteristics manner in all the treatments (Table 3). The minimum radial growth (3.46) was recorded in 1000ppm salicylic acid. The next higher radial growth (4.48) was recorded in 200ppm isonicotinamide (Plate No. 2 and 3). The next higher radial growth (6.75) was observed in 1000ppm isonicotinamide followed by 600ppm salicylic acid (6.94),

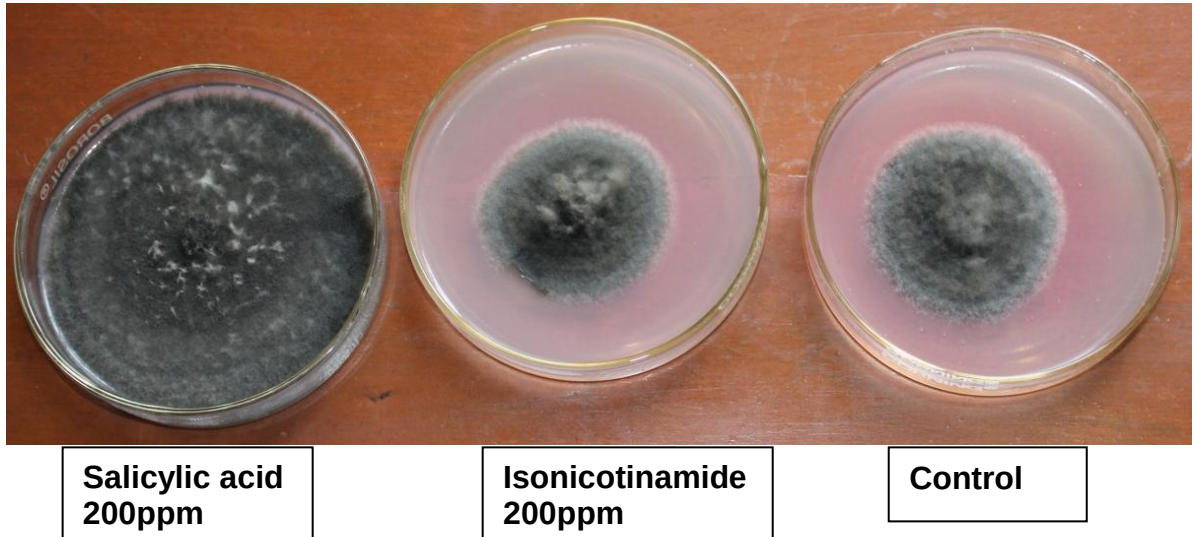


Plate 2:- Radial growth of *Rhizoctonia bataticola* as influenced by chemical elicitors.

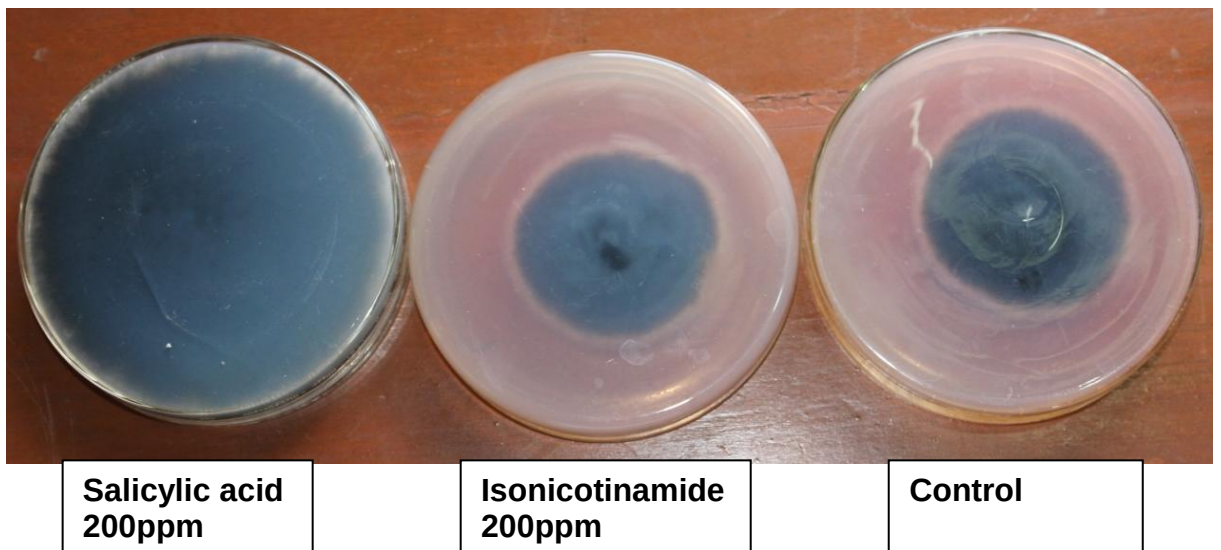


Plate 3:- Substrate colour of *Rhizoctonia bataticola* as influenced by chemical elicitors.



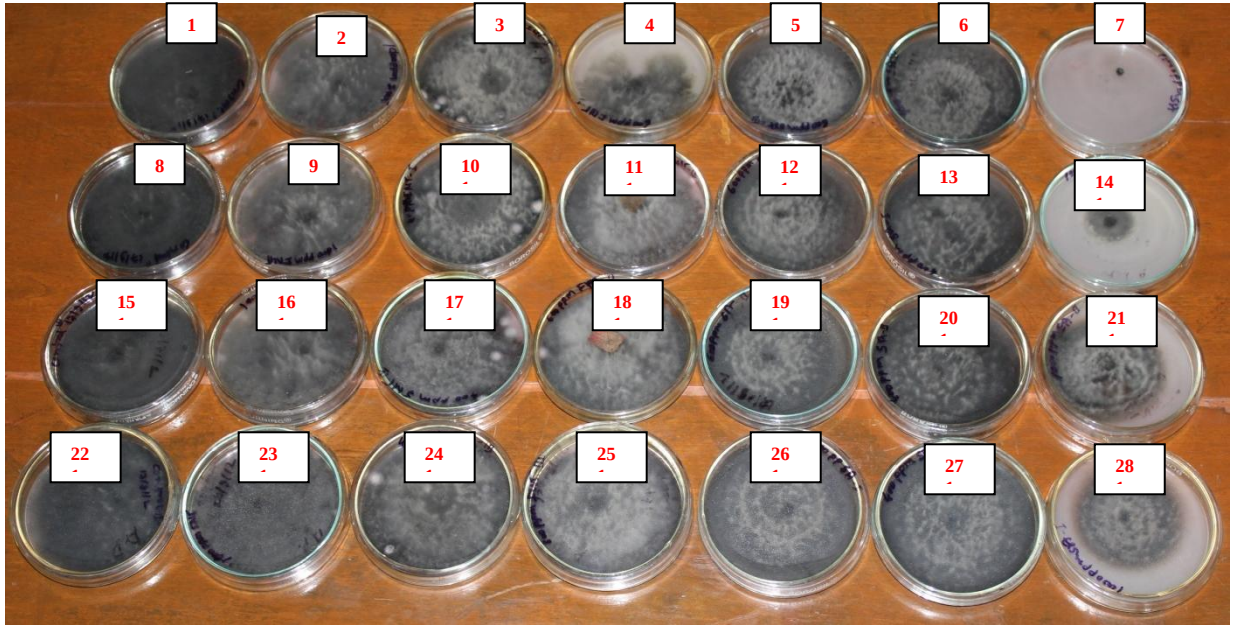
Control

**Salicylic acid
400ppm**

**Salicylic acid
600ppm**

**Salicylic acid
1000ppm**

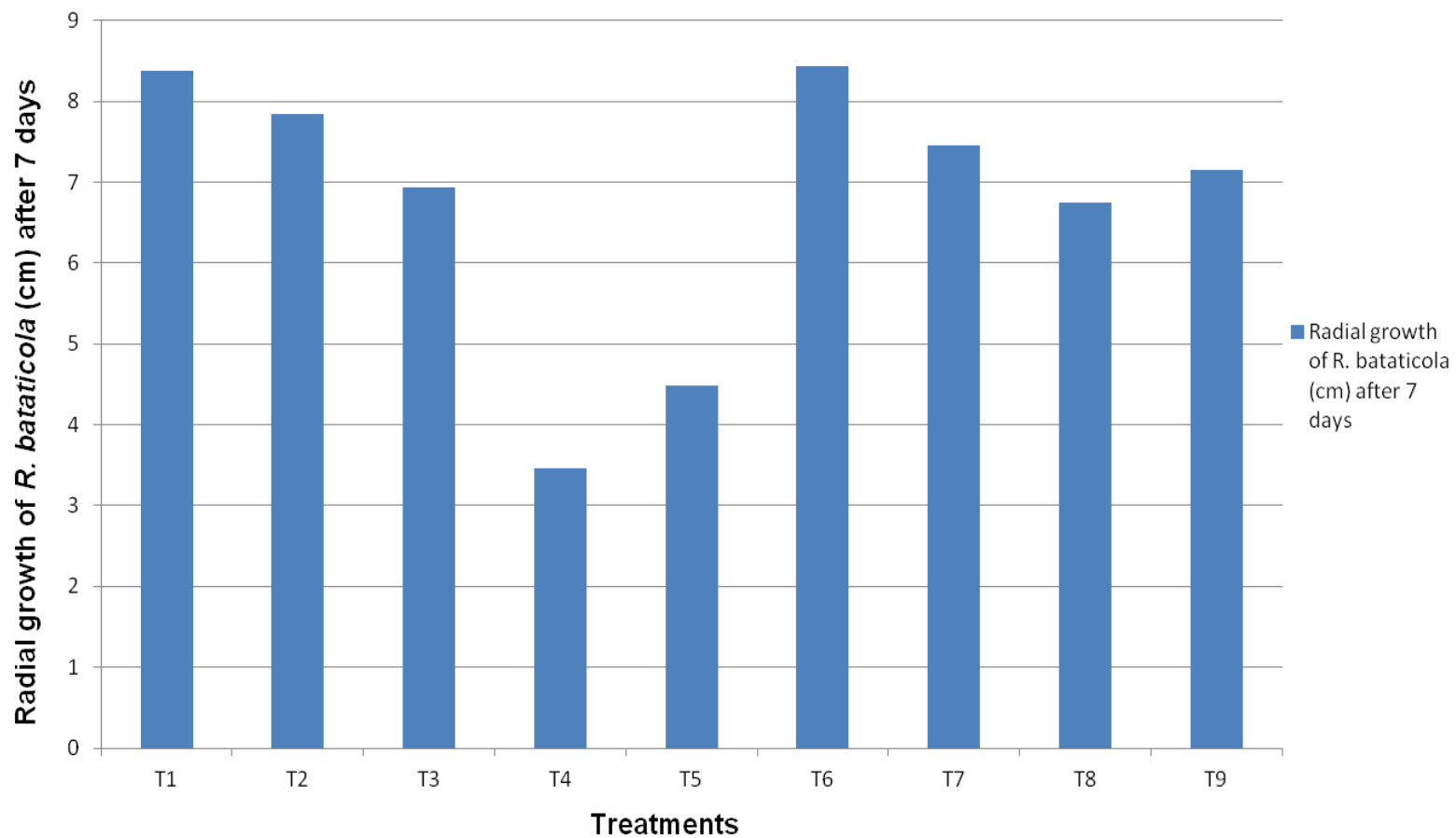
Plate 4:- Radial growth of *Rhizoctonia bataticola* as influenced by chemical elicitors.



1. Control 2. Isonicotinamide 1000ppm 3. Isonicotinamide 400ppm
4. Isonicotinamide 600ppm 5. Salicylic acid 600ppm 6. Salicylic acid 400ppm
7. Salicylic acid 1000ppm 8. Control 9. Isonicotinamide 1000ppm
10. Isonicotinamide 400ppm 11. Isonicotinamide 600ppm 12. Salicylic acid 600ppm
13. Salicylic acid 400ppm 14. Salicylic acid 1000ppm 15. Control
16. Isonicotinamide 1000ppm 17. Isonicotinamide 400ppm 18. Isonicotinamide 600ppm
19. Salicylic acid 600ppm 20. Salicylic acid 400ppm 21. Salicylic acid 1000ppm
22. Control 23. Isonicotinamide 1000ppm 24. Isonicotinamide 400ppm
25. Isonicotinamide 600ppm 26. Salicylic acid 600ppm 27. Salicylic acid 400ppm
28. Salicylic acid 1000ppm

Plate 5:- Radial growth of *Rhizoctonia bataticola* as influenced by chemical elicitors.

Fig:-3 Radial growth of *R. bataticola* as influenced by chemical elicitors.



control (7.15) 600ppm isonicotinamide (7.45) which were statistically at par. Still the next higher radial growth (6.94) was recorded in 600ppm salicylic acid followed by control (7.15), 600ppm isonicotinamide (7.45) and 400ppm salicylic acid (7.84). The maximum radial growth was recorded in 400ppm isonicotinamide (8.43) which was statistically at par with earlier set of 400ppm salicylic acid (7.84) and 200ppm salicylic acid (8.38) (Plate No. 4).

Thus the use of 1000ppm salicylic acid was most effective among all the treatments when supported minimum radial growth of the pathogen (Plate No. 5).

Table 3 : Radial growth of *Rhizoctonia bataticola* as influenced by chemical elicitors.

Treatments	Treatments description	Radial growth of <i>R. bataticola</i> (cm) after 7 days
T1	200ppm salicylic acid	8.38
T2	400ppm salicylic acid	7.84
T3	600ppm salicylic acid	6.94
T4	1000ppm salicylic acid	3.46
T5	200ppm isonicotinamide	4.48
T6	400ppm isonicotinamide	8.43
T7	600ppm isonicotinamide	7.45
T8	1000ppm isonicotinamide	6.75
T9	Control	7.15
	Sem±	0.32
	CD	0.92

4.8 Incidences of dry root rot in chickpea under the influence of chemical elicitors in poly house.

The data presented in table 4 on the effect of two chemical elicitors in the chickpea for the control of dry root rot disease in poly house revealed that significantly less disease incidence was recorded by all the chemical elicitors on the control.

Fig:-4 Incidence of dry root rot in chickpea under the influence of chemical elicitors in the poly house.

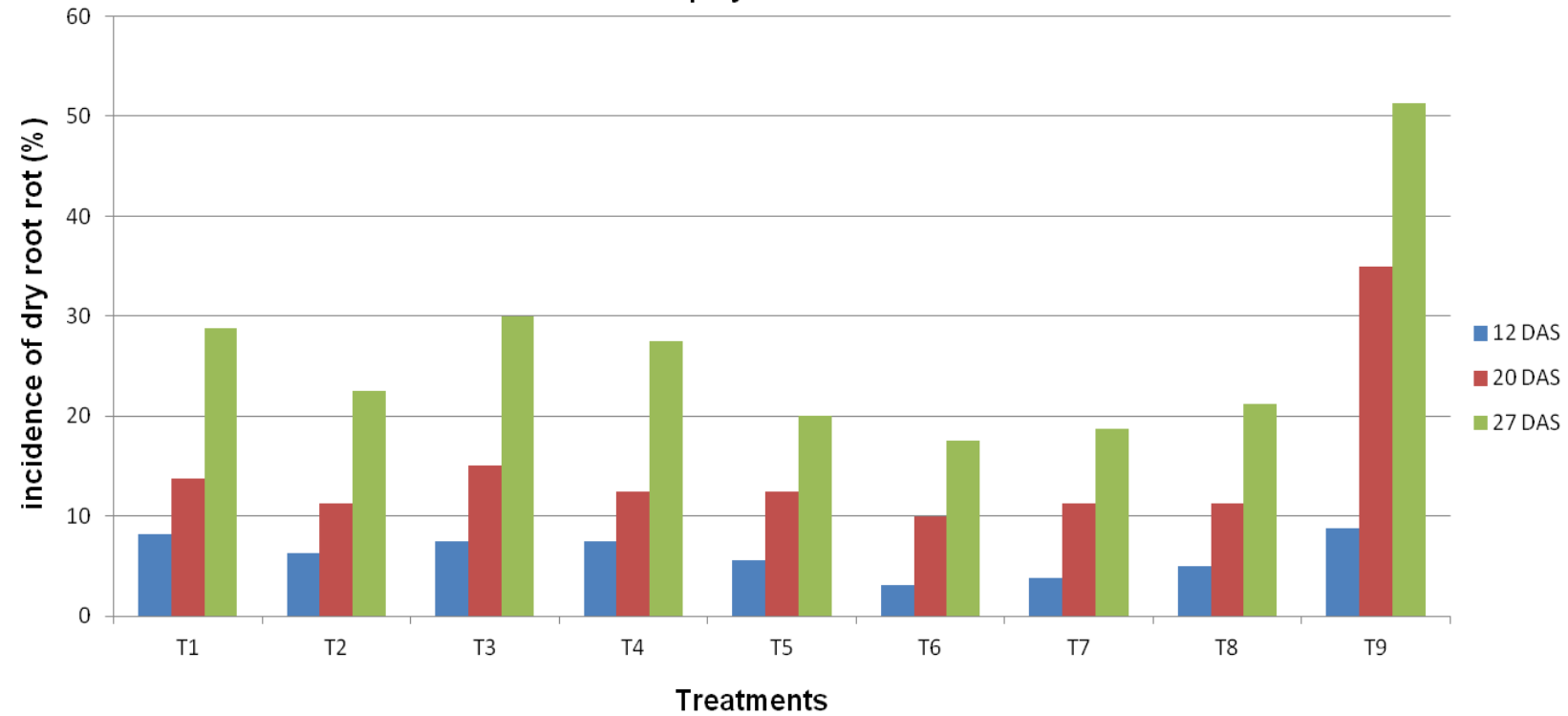


Table 4: Incidence of dry root rot in chickpea under the influence of chemical elicitors in the poly house.

Treatments	Treatment Description	Mean incidence of dry root rot (%)		
		12 DAS	20 DAS	27 DAS
T1	ST with 200ppm SA	8.13(16.42)	13.75(21.70)	28.75(32.37)
T2	ST with 400ppm SA	6.25(14.30)	11.25(19.52)	22.50(28.28)
T3	ST with 200ppm INT	7.50(15.68)	15.00(22.64)	30.00(33.17)
T4	ST with 400ppm INT	7.50(15.79)	12.50(20.61)	27.50(31.51)
T5	ST with 200ppm SA and FS of SA with 200ppm just after germination of the plants	5.63(13.34)	12.50(20.61)	20.00(26.48)
T6	ST with 400ppm SA and FS of SA with 400ppm just after germination of the plants	3.13(10.05)	10.00(18.14)	17.50(24.68)
T7	ST with 200ppm INT and FS of INT with 200ppm just after germination of the plants	3.75(11.01)	11.25(19.23)	18.75(25.39)
T8	ST with 400ppm INT and FS of INT with 400ppm just after germination of the plants	5.00(12.92)	11.25(19.52)	21.25(27.42)
T9	Control	8.75(17.06)	35.00(36.22)	51.25(45.72)
	Sem±	1.30	1.57	1.59
	CD	2.43	2.74	4.60

*The data transformed by Angular transformation.

i. Incidence of dry root rot 12 DAS:

The minimum disease incidence (10.05) was observed in application of seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm just after germination of the plants which was statistically identical to seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants (11.01). The next higher disease incidence (11.01) was observed in seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants which was followed by seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm just after germination of the plants (12.92) which were statistically at par to each other. The next higher disease incidence (12.92) was recorded in seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm just after germination of the plants.

A disease incidence of (14.30) was recorded in seed treatment with 400ppm salicylic acid which was statistically similar to seed treatment with 200ppm isonicotinamide (15.68), seed treatment with 400ppm isonicotinamide (15.79) and seed treatment with 200ppm salicylic acid (16.42).

The maximum disease incidence (17.06) was recorded in control which was statistically higher than all the treatments.

Thus salicylic acid at 400ppm applied as seed treatment along with its foliar application just after germination of the plants resulted in minimum incidence of dry root rot and thus has shown slight resistance against the disease.

The next effective treatment was salicylic acid at 200ppm applied as seed treatment coupled with foliar application just after germination of the plants.

ii. Incidence of dry root rot 20 DAS:

The minimum disease incidence (18.14) was observed in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid

with 400ppm just after germination of the plants which was similar to seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants (19.23), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm just after germination of the plants and seed treatment with 400ppm salicylic acid (19.52), seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants and seed treatment with 400ppm isonicotinamide (20.61). the next higher disease incidence (19.52) observed in seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm just after germination of the plants followed by seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm just after germination of the plants and seed treatment with 400ppm salicylic acid (19.52), seed treatment with 400ppm isonicotinamide and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants (20.61), seed treatment with 200ppm salicylic acid (21.70) which were statistically at par. still higher incidence was observed in seed treatment with 200ppm isonicotinamide (22.64) which was statistically non-significant with the disease incidence seed treatment with 400ppm isonicotinamide and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants (21.70) in seed treatment with 200ppm salicylic acid.

The maximum disease incidence (36.22) was recorded in control which was statistically higher than all the treatment.

Thus salicylic acid at 400ppm applied as seed treatment along with its foliar application just after germination of the plants resulted in minimum incidence of dry root rot and thus has shown slight resistance against the disease.

The next effective treatment was salicylic acid at 200ppm applied as seed treatment coupled with foliar application just after germination of the plants.

iii. Incidence of dry root rot 27 DAS:

The minimum disease incidence (24.68) was recorded in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm just after germination of the plants which was statistically identical to seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants (25.39), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm just after germination of the plants (26.48), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm just after germination of the plants (27.42) and seed treatment with 400ppm salicylic acid (28.28). The next higher disease incidence (27.42) was observed in seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm just after germination of the plants followed by seed treatment with 400ppm salicylic acid (28.28) and seed treatment with 400ppm isonicotinamide (31.51) still the next higher disease incidence (28.28) was observed in seed treatment with 400ppm salicylic acid followed by seed treatment with 400ppm isonicotinamide (31.51) and seed treatment with 200ppm salicylic acid (32.37) which were statistically at par. A disease incidence of (33.17) was observed in seed treatment with 200ppm isonicotinamide which was statistically at par with the earlier set of seed treatment with 400ppm isonicotinamide (31.51) and seed treatment with 200ppm salicylic acid (32.37). The maximum disease incidence (45.72) was observed in control which was statistically higher than all the treatments.

Thus salicylic acid at 400ppm applied as seed treatment along with its foliar application just after germination of the plants resulted in minimum incidence of dry root rot and thus has shown slight resistance against the disease.

The next effective treatment was salicylic acid at 200ppm applied as seed treatment coupled with foliar application just after germination of the plants.

Thus the seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm just after germination of the plants showed minimum disease incidence and induction of resistance against the disease at variable plant age 12, 20 and 27 DAS.

4.9 Influence of chemical elicitors on germination percent of chickpea in the poly house.

The data presented in table 5 on the effect of 2 chemical elicitors in the chickpea for the control of dry root rot disease in poly house revealed non-significant differences in germination (%) by all the chemical elicitors over the control.

Among chemical elicitors the minimum germination percent (44.28) was recorded in control which was statistically identical to seed treatment with 200ppm salicylic acid (44.71) and seed treatment with 200ppm isonicotinamide (46.44). The maximum germination percent (56.89) was observed in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm just after germination of the plants which was statistically at par with the earlier set of seed treatment with 400ppm salicylic acid (52.29) and seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm just after germination of the plants (52.29), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm just after germination of the plants (53.75), seed treatment with 400ppm isonicotinamide (54.52) and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants (56.06).

Thus the data revealed that there were statistically non-significant difference between all the treatments but the highest germination was achieved by the seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm just after germination of the plants followed by seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants.

Fig:-5 Germination of chickpea seeds under the influence of chemical elicitors in the poly house at 5 DAS.

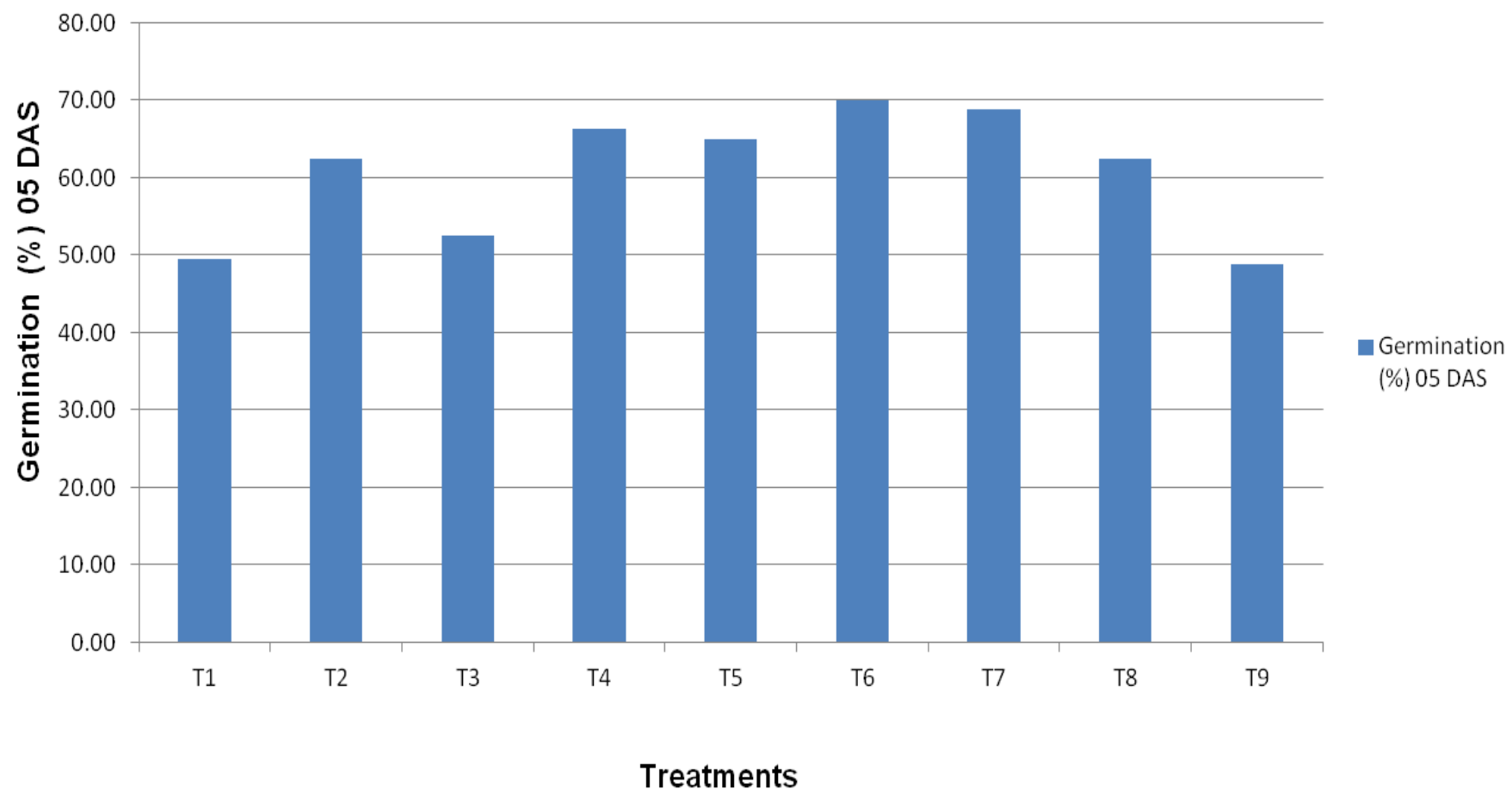


Table 5: influence of chemical elicitors on germination percent of chickpea at 05 DAS in the poly house

Treatments	Treatment Description	Germination (%) 05 DAS
T1	ST with 200ppm SA	49.50(44.71)*
T2	ST with 400ppm SA	62.50(52.29)
T3	ST with 200ppm INT	52.50(46.44)
T4	ST with 400ppm INT	66.25(54.52)
T5	ST with 200ppm SA and FS of SA with 200ppm just after germination of the plant	65.00(53.75)
T6	ST with 400ppm SA and FS of SA with 400ppm just after germination of the plant	70.00(56.89)
T7	ST with 200ppm INT and FS of INT with 200ppm just after germination of the plant	68.75(56.06)
T8	ST with 400ppm INT and FS of INT with 400ppm just after germination of the plant	62.50(52.29)
T9	Control	48.75(44.28)
	Sem±	1.81
	CD	5.23

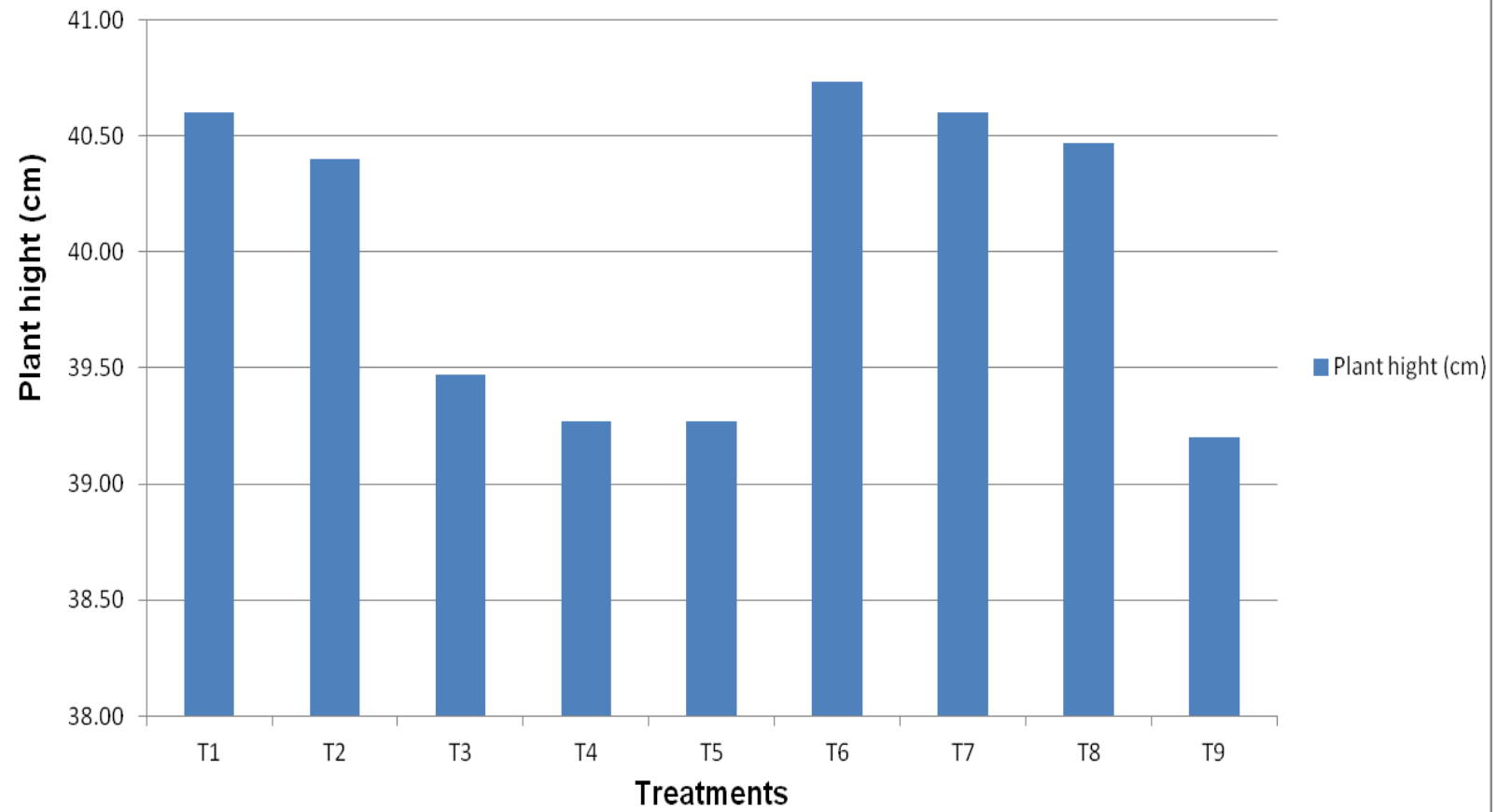
*The data transformed by Angular transformation.

4.10 Height of chickpea plants as influenced by chemical elicitors

Height of plants (cm) was measured with the help of scale. The data on this growth attributed character was recorded at harvesting time. The data showed that all the treatments showed non- significant variation in plant height.

The minimum plant height (39.20) was recorded in control followed by seed treatment with 400ppm isonicotinamide (39.27), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (39.27), seed treatment with 200ppm isonicotinamide (39.47), seed treatment with 400ppm salicylic acid (40.40),

Fig:-6 Height of chickpea plants as influenced by chemical elicitors)



seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (40.47), seed treatment with 200ppm salicylic acid (40.60), seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (40.60), and seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS (40.73).

Thus the data revealed that there were statistically non-significant difference between all the treatments but the seed treatment with salicylic acid 200ppm resulted in maximum of plant.

Table 6: Height of chickpea plants as influenced by chemical elicitors

Treatments	Treatment description	Plant height (cm)
T1	ST with 200ppm SA	40.60
T2	ST with 400ppm SA	40.40
T3	ST with 200ppm INT	39.47
T4	ST with 400ppm INT	39.27
T5	ST with 200ppm SA and FS of SA with 200ppm at 30 DAS	39.27
T6	ST with 400ppm SA and FS of SA with 400ppm at 30 DAS	40.73
T7	ST with 200ppm INT and FS of INT with 200ppm at 30 DAS	40.60
T8	ST with 400ppm INT and FS of INT with 400ppm at 30 DAS	40.47
T9	Control	39.20
	Sem±	1.14
	CD	NS

4.11 Number of branches per plants as influenced by chemical elicitors

The data of number of branches per plot at the time of harvesting of chickpea is given in table 7. The number of branches enhanced tremendously with the increment in the growth and development of plant up to the harvesting.

The data indicated that the minimum number of branches (10.87) was observed in control and seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (10.87) which was statistically identical to treatment with 400ppm salicylic acid (10.93), seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (11.13), seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (11.20), seed treatment with 200ppm isonicotinamide (11.27), seed treatment with 400ppm isonicotinamide (11.67). The maximum number of branches (12.07) was observed in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS which was statistically at par with the earlier set of seed treatment with 200ppm isonicotinamide (11.27), seed treatment with 400ppm isonicotinamide (11.67) and seed treatment with 200ppm salicylic acid (11.80).

Although there were statistically non-significant difference between all the treatments yet the application of salicylic acid 400ppm as seed treatment coupled with foliar application at 30 DAS resulted in maximum number of branches/plant.

Fig:-7 Number of branches per plants as influenced by chemical elicitors.

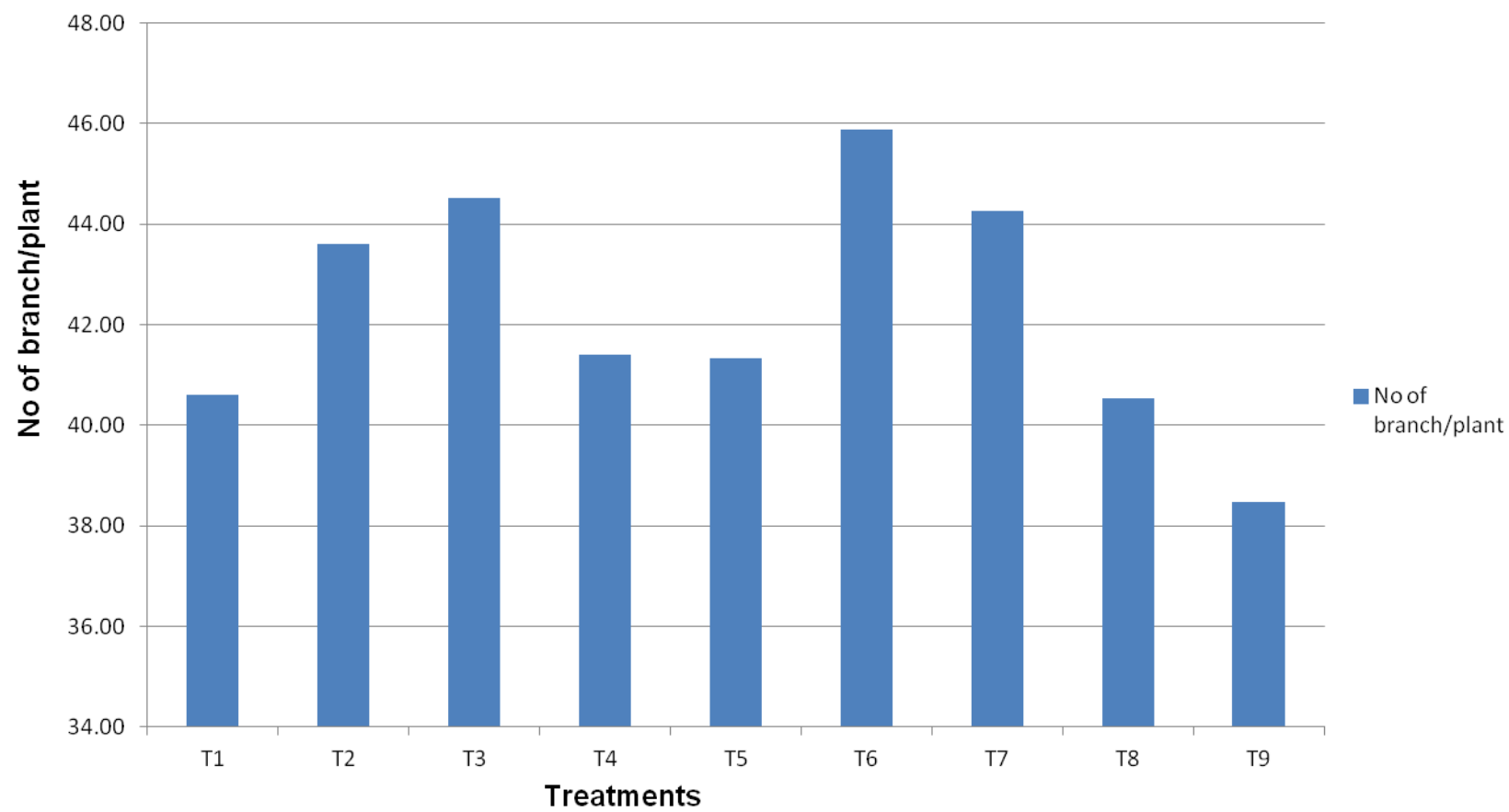


Table 7: Number of branches per plant as influenced by chemical elicitors

Treatments	Treatment description	Number of branches/plant
T1	ST with 200ppm SA	11.80
T2	ST with 400ppm SA	11.67
T3	ST with 200ppm INT	11.27
T4	ST with 400ppm INT	11.67
T5	ST with 200ppm SA and FS of SA with 200ppm at 30 DAS	10.87
T6	ST with 400ppm SA and FS of SA with 400ppm at 30 DAS	12.07
T7	ST with 200ppm INT and FS of INT with 200ppm at 30 DAS	11.13
T8	ST with 400ppm INT and FS of INT with 400ppm at 30 DAS	11.20
T9	Control	10.87
	Sem$\pm$	0.28
	CD	NS

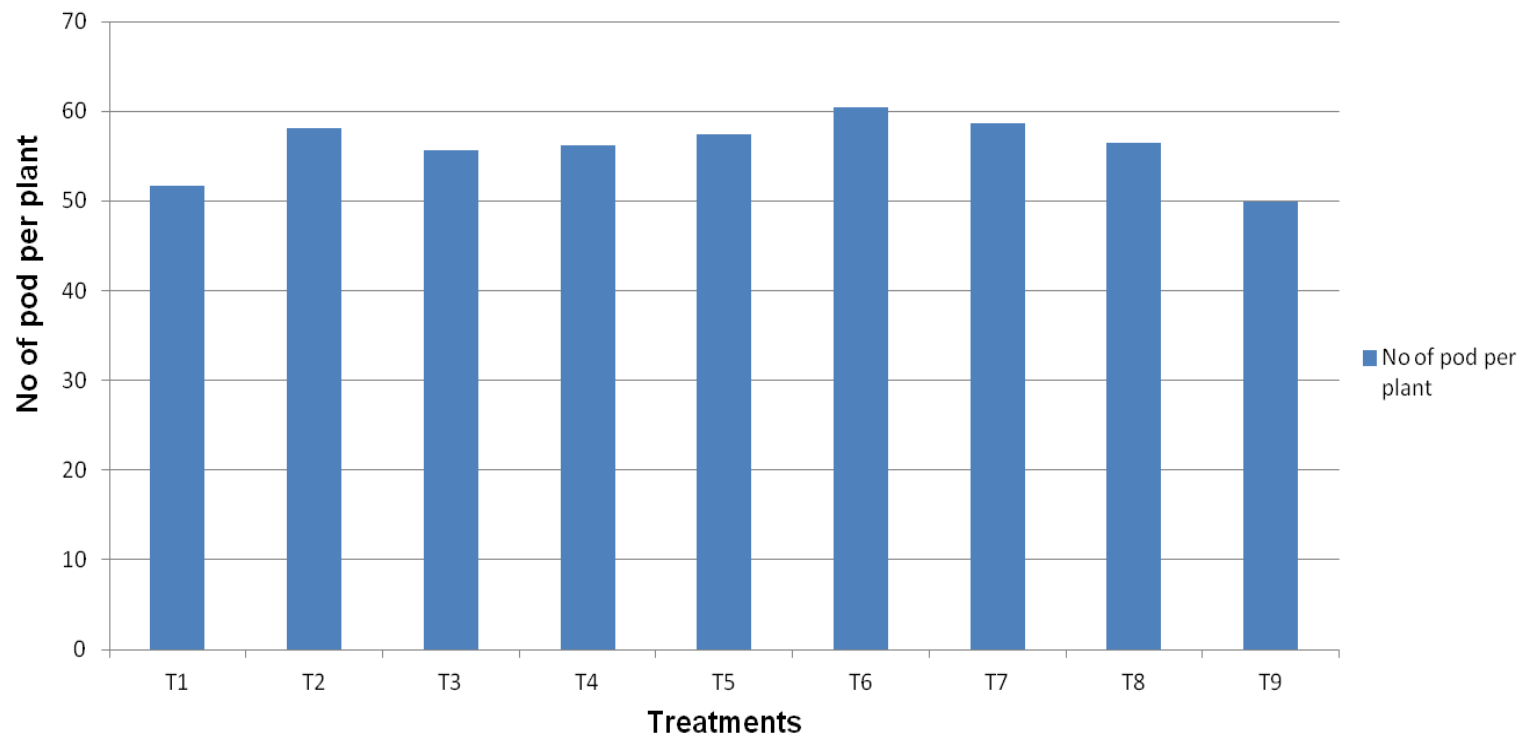
4.12 Number of pods per plants as influenced by chemical elicitors

The number of pods per plant is an important yield attributing character ultimately influences the grain yield of any crop. The data on mean number of the pods per plant as affected by the different treatments is presented in table 8. The number of pods per plant was non-significantly affected by all the treatments.

The minimum number of pods (49.87) was recorded in control which was statistically identical to seed treatment with 200ppm salicylic acid (51.73), seed treatment with 200ppm isonicotinamide (55.60), seed treatment with 400ppm isonicotinamide (56.13), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (56.53), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (57.47) and seed treatment with 400ppm salicylic acid (58.07). The higher number of pods was recorded in seed treatment with 200ppm salicylic acid (51.73) followed by seed treatment with 200ppm isonicotinamide (55.60), seed treatment with 400ppm isonicotinamide (56.13), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (56.53), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (57.47), seed treatment with 400ppm salicylic acid (58.07) and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (58.60).

The maximum number of pods was recorded in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS (60.40) which was statistically higher than all the treatments except seed treatment with 200ppm isonicotinamide (55.60), seed treatment with 400ppm isonicotinamide (56.13), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (56.53), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (57.47), seed treatment with 400ppm salicylic acid (58.07) and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (58.60).

Fig:-8 Number of pods par plants as influenced by chemical elicitors.



Thus the application of salicylic acid 400ppm as seed treatment coupled with foliar application at 30 DAS resulted in maximum number of pods/plant.

Table 8: Number of pods per plant as influenced by chemical elicitors

Treatments	Treatment description	No of pods per plant
T1	ST with 200ppm SA	51.73
T2	ST with 400ppm SA	58.07
T3	ST with 200ppm INT	55.60
T4	ST with 400ppm INT	56.13
T5	ST with 200ppm SA and FS of SA with 200ppm at 30 DAS	57.47
T6	ST with 400ppm SA and FS of SA with 400ppm at 30 DAS	60.40
T7	ST with 200ppm INT and FS of INT with 200ppm at 30 DAS	58.60
T8	ST with 400ppm INT and FS of INT with 400ppm at 30 DAS	56.53
T9	Control	49.87
	Sem$\pm$	2.89
	CD	NS

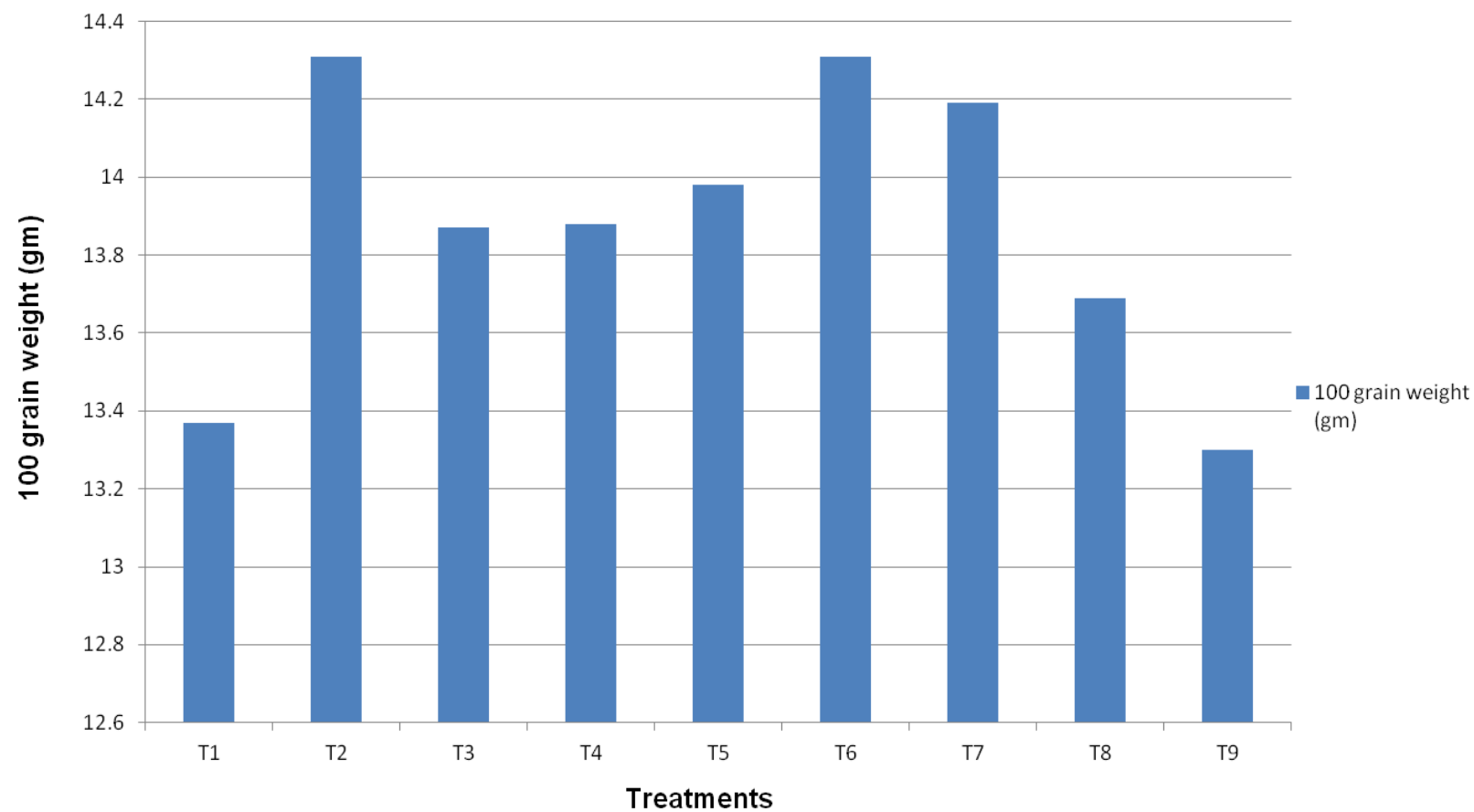
4.13 100 grain weight as influenced by chemical elicitors

The data given in table 9 showed that only seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS (14.31) influenced 100 seed weight of chickpea maximum but it was non-significant. The treatments were at par with each other in respect of this character.

The minimum 100 grain weight (13.30) found in control which was identical to seed treatment with 200ppm salicylic acid (13.37), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (13.69), seed treatment with 200ppm isonicotinamide (13.78), seed treatment with 400ppm isonicotinamide (13.88) and seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (13.98). The higher 100 grain weight was recorded in seed treatment with 200ppm salicylic acid (13.37) which was similar to seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (13.69), seed treatment with 200ppm isonicotinamide (13.78), seed treatment with 400ppm isonicotinamide (13.88), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (13.98) and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (14.19).

The maximum 100 grain weight (14.31) was recorded in seed treatment with 400ppm salicylic acid and seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS (14.31) which was statistically higher than all the treatments except seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (13.69), seed treatment with 200ppm isonicotinamide (13.78), seed treatment with 400ppm isonicotinamide (13.88), seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (13.98) and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (14.19).

Fig:-9 100 grain weights as influenced by chemical elicitors.



The best treatment when enhanced the 100 grain weight was salicylic acid 400ppm coupled with its foliar application at 30DAS.

Table 9: 100 grain weight as influenced by chemical elicitors

Treatments	Treatment description	100 grain weight
T1	ST with 200ppm SA	13.37
T2	ST with 400ppm SA	14.31
T3	ST with 200ppm INT	13.87
T4	ST with 400ppm INT	13.88
T5	ST with 200ppm SA and FS of SA with 200ppm at 30 DAS	13.98
T6	ST with 400ppm SA and FS of SA with 400ppm at 30 DAS	14.31
T7	ST with 200ppm INT and FS of INT with 200ppm at 30 DAS	14.19
T8	ST with 400ppm INT and FS of INT with 400ppm at 30 DAS	13.69
T9	Control	13.30
	Sem±	0.28
	CD	NS

4.14 Yield of chickpea in the influence of chemical elicitors

The grain yield is the ultimate things for assessment of treatment in any agronomical investigation. The data recorded on grain yield in kg/plot was converted in to grain yield in q/ha.

The minimum yield (10.22) was observed in control which was statistically identical to seed treatment with 400ppm isonicotinamide (10.56), seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (10.60), seed treatment with 200ppm salicylic acid (12.35), seed treatment with 200ppm isonicotinamide (13.87) and seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (14.54). The higher grain yield was recorded in seed treatment with 400ppm isonicotinamide (10.56) followed by seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS (10.60), seed treatment with 200ppm salicylic acid (12.35), seed treatment with 200ppm isonicotinamide (13.87) and seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (14.54) and seed treatment with 400ppm salicylic acid (14.60).

The maximum grain yield was recorded in seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS (15.86) which was statistically higher than all the treatments except seed treatment with 200ppm salicylic acid (12.35), seed treatment with 200ppm isonicotinamide (13.87) and seed treatment with 200ppm salicylic acid and foliar application of salicylic acid with 200ppm at 30 DAS (14.54) and seed treatment with 400ppm salicylic acid (14.60) and seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS (15.25).

Thus the data presented in table 4.10 revealed that all the treatments resulted in non-significant differences in the grain yield of chickpea. However, the maximum yield was obtained by the application of salicylic acid at 400ppm as seed treatment alone and coupled with its foliar application at 30DAS which was followed by seed treatment with isonicotinamide 200ppm coupled with foliar application 200ppm at 30DAS.

Fig:-10 Yield of chickpea under in the influence of chemical elicitors (q/ha).

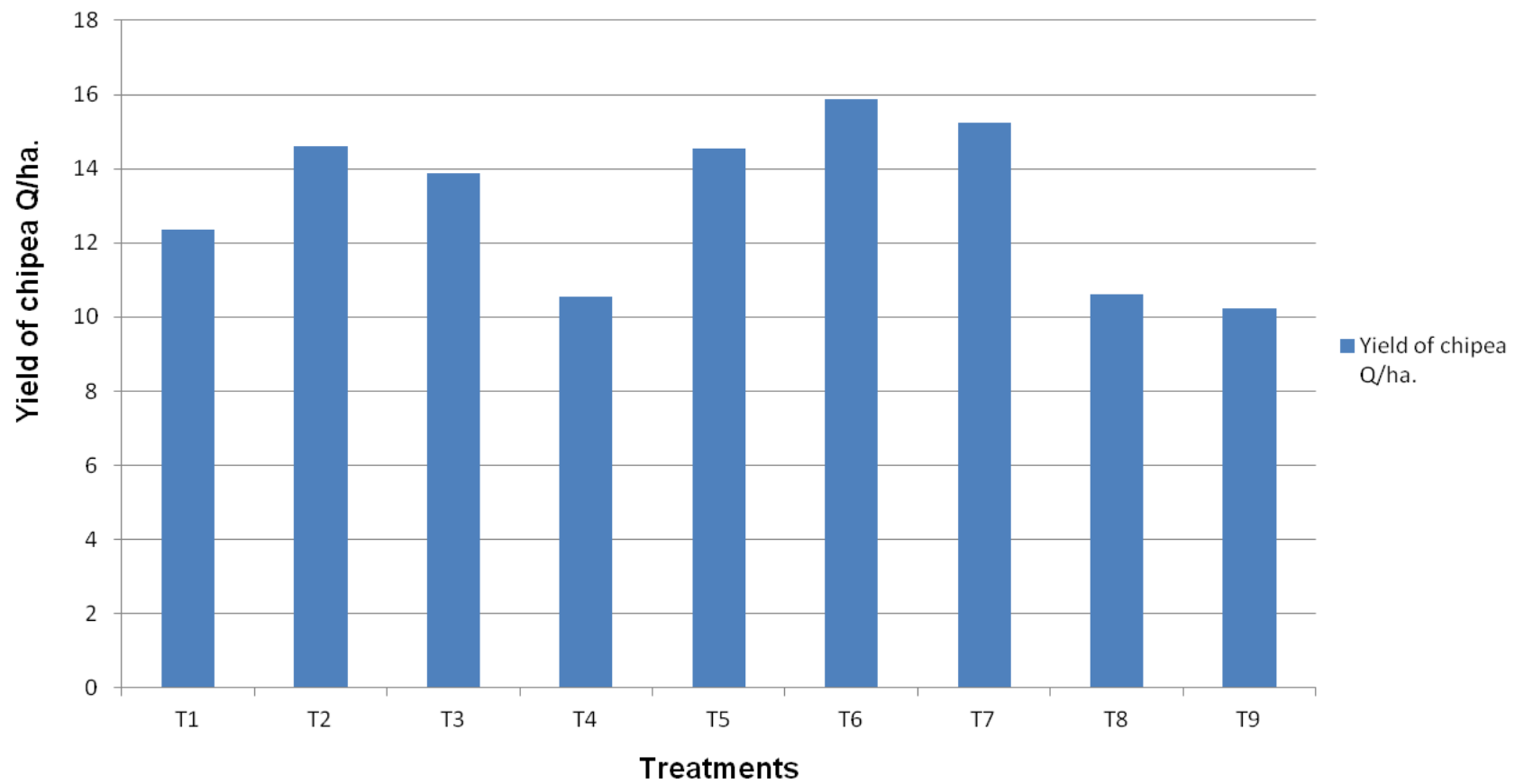


Table 10: Yield of chickpea in the influence of chemical elicitors

Treatments	Treatment description	Yield of chickpea (q/ha)
T1	ST with 200ppm SA	12.35
T2	ST with 400ppm SA	14.60
T3	ST with 200ppm INT	13.87
T4	ST with 400ppm INT	10.56
T5	ST with 200ppm SA and FS of SA with 200ppm at 30 DAS	14.54
T6	ST with 400ppm SA and FS of SA with 400ppm at 30 DAS	15.86
T7	ST with 200ppm INT and FS of INT with 200ppm at 30 DAS	15.25
T8	ST with 400ppm INT and FS of INT with 400ppm at 30 DAS	10.60
T9	Control	10.22
	Sem$\pm$	1.44
	CD	NS

CHAPTER- 5

DISCUSSION

The experimental findings on various aspects of dry root rot of chickpea during *Rabi* 2011-12 i.e. **evaluation of SAR inducing chemicals against *Rhizoctonia bataticola* in vitro**, evaluation of SAR inducing chemicals for control of dry root rot in poly house and for its control in the field at different concentrations are discussed here under.

Major losses in chickpea yield are attributed mainly due to soil borne pathogens. The crop suffers from a number of soil borne pathogens like dry root rot *Rhizoctonia bataticola*, collar rot *Sclerotium rolfsii* and wilt – *Fusarium oxysporum* f.sp. *ciceris*. Among these wilt caused by *Fusarium oxysporum* Schl. Emend, Snyder and Hans f. sp. *Ciceris* (Padwick) Snyder and Hans is common and important throughout the world (Anonymous 1982.; Gupta *et al.*, 1987). Dry root rot incited by *Rhizoctonia bataticola* (Taub), Butler, is another serious disease at early crop stage.

5.1 Symptomatology

Critical and close observations on the symptomatology of the disease *in vitro* as well as *in vivo* helped in the identification of the causal organism. Drying of the plants appears suddenly in the field at flowering and podding stages of the crop growth. There is no drooping which is commonly seen in the plants affected by wilt. The leaves and stems of affected plants are straw-colored. The roots are usually dry, unless the soil is wet. The tap root is quite brittle, shows shredding of the bark and can be broken easily. Minute dark black sclerotial bodies can be seen with the aid of a 10 X hand lens on the surface of the root, as well as in the pith.

The disease appears suddenly when ambient temperatures are between 25 and 30°C.

Pathogen: *Rhizoctonia bataticola* (Taub) Butler=*Macrophomina phaseolina* (Maubl.) Ashby

5.2 Isolation of the causal organism

The pathogen *Rhizoctonia bataticola* was isolated from root region of the diseased plants by time segment method (Rangashwamy, 1972) and purified by hyphal tip method and maintained on potato dextrose agar slants.

5.3 Characterization of the pathogen

The morphological and cultural characteristics and formation of the sclerotia were the principle characters to identify the pure cultures of *R. bataticola*. The characters were compared with those described by Ashby (1927) and the fungus was identified as *R. bataticola*.

The fungus lacks fruiting bodies and spores. The mycelium is light-brown, thick in which black sclerotia are formed. Sclerotia are variable in form, small loosely connected by mycelial threads.

5.4 Multiplication of fungal inoculum

The fungal inoculum was multiplied on sterilized sand maize meal medium in 1000 ml flasks containing 400 g in each flask. A loop full of fungus from actively growing culture was inoculated into each flask and incubated at room temperature for 15 days. The fungal culture multiplied on sand maize meal medium was used for inoculation to the soil.

5.5 Incidence of dry root rot in chickpea under the influence of chemical elicitors

Two chemical elicitors i.e. salicylic acid and isonicotinamide were applied on chickpea variety JG-62 at 200ppm and 400ppm as seed treatment alone and in combination with their foliar application 30DAS in a field experiment to see their effect on imparting resistance against dry root rot (as a reaction in the incidence of the disease).

The data on incidence of dry root rot (%) 30, 60 and 90 DAS have been presented in table 1. The data on disease incidence 30 DAS among different treatments showed significant differences.

Salicylic acid at 400ppm applied as seed treatment along with its foliar application at 30 DAS, 60 DAS and 90 DAS, respectively resulted in

minimum incidence of dry root rot (8.25), (15.37) and (23.44) and thus has shown slight resistance against the disease. The next effective treatment was seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30DAS, 60DAS and 90DAS (9.99), (15.68) and (25.11), respectively. Gaffney *et al.* (1993) reported that the present paper reports that seed treatment with salicylic acid and Bion induces systemic resistance in chickpea seedlings against *Fusarium* wilt., Delaney *et al.*, (1995) reported that salicylic acid is a natural phenolic compound present in many plants, and an important component in the signal transduction pathway involved in local and systemic resistances to pathogens., Hijwegen and Verhaar (1995) reported that isonicotinic acid decreased powdery mildew infections in cucumber., Hijwegen *et al.* (1996) reported that INA decreased powdery mildew development in roses. Matta (2001) reported that SAR (systemic acquired resistance) is a type of induced resistance that follows necrogenous infections and is characterized by accumulation of salicylic acid and increased synthesis of pathogenesis related proteins., Raju *et al.* (2008) reported that induction of some defense related enzymes and phenolics in roots and shoots of two different genotypes of chickpea which were susceptible (L 550) and resistant (ICCV 10) to wilt disease treated with salicylic acid, spermine (Spm), SA+Spm and *Fusarium oxysporum* f. sp. *ciceri*. Higher levels of polyphenol oxidase (PPO), phenylalanine ammonia-lyase (PAL), 1, -3-glucanase (PR-2) and phenolics were observed in roots and shoots of resistant cultivar than that of susceptible cultivar on treatment with elicitors and pathogen. Presently also the use of salicylic acid and isonicotinamide both a seed treatment and foliar application resulted in reduction of disease incidence. These findings are in conformity with the present observations.

5.6 Influence of chemical elicitors on germination percent of chickpea at 10 DAS

Two chemical elicitors i.e. salicylic acid and isonicotinamide were applied on chickpea variety JG-62 at 200ppm and 400ppm as seed treatment

alone in a field experiment to see their effect on enhancing the germination (%) of chickpea.

The data of chickpea germinated plants in the plots as recorded at 10 DAS were subjected to statistical analysis. The analysis of variance computed from these data is given in table 2. Two chemical elicitors i.e. salicylic acid and isonicotinamide exhibited that their different concentration did not produce any significant difference on the germination percent in the plots of chickpea at 10DAS.

Although the highest germination was achieved by the seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS followed by seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm at 30 DAS yet the differences among the treatments were statistically non-significant.

5.7 Radial growth of *Rhizoctonia bataticola* as influenced by chemical elicitors

For studying the reaction of 2 chemical elicitors against *R. bataticola* poisoned food technique was followed (Nene and Thapliyal, 1993). Four doses of chemical elicitors were used exhibiting an increasing trend of colony diameter when the concentration of the chemical elicitor decreased than the normal dose.

Thus the use of 1000ppm salicylic acid was most effective among all the treatments when supported minimum radial growth of the pathogen. Ali *et al.* (2009) reported that efficacy of some plant resistance elicitors viz: chitosan (CHI), salicylic acid (SA) and hydroquinone (HQ) in comparing to the fungicide Rhizolex T-50 as seed treatments showed significant reduction in the fungal growth *in vitro.*, Amal (2009) reported that the inhibitory effect of twelve chemicals from different groups of known resistance inducers to the growth of bean seed-borne fungi was evaluated, *in vitro*, at concentrations of 1, 5, 10 and 20 mM. the fungal growth was negatively affected by increasing concentrations of the tested inducer chemicals. benzoic, salicylic and sorbic acids completely suppressed fungal growth at 10 mM., Abdel-Monaim and Ismail (2010) reported that the four antioxidant compounds (coumaric acid,

citric acid, propylgalate and salicylic acid each at 100 and 200 ppm) were evaluated for their *in vitro* and *in vivo* against to *Fusarium* pathogenic isolates caused root rot and wilt diseases in pepper plants. All chemicals reduced spore formation in both *Fusarium species* at 100 or 200ppm to a great extent and the inhibitory effect of antioxidants increased with increasing their concentrations. These findings support the present observations with salicylic acid and isonicotinamide with *R. bataticola*.

5.8 Incidence of dry root rot in chickpea under the influence of chemical elicitors in poly house

The data presented in table 4 as the effect of two chemical elicitors in the chickpea for the control of dry root rot disease in poly house revealed that significantly less disease incidence was recorded by all the chemical elicitors than the control.

The seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm just after germination of the plants showed minimum disease incidence and induction of resistance against the disease at variable plant age 12, 20 and 27 DAS. Kubo and Sato (2002) proved that salicylic acid (SA), a chemical inducer of SAR, and SAR gene expression were induced by high-temperature treatments in greenhouse-grown cucumbers, providing evidence for the existence of SAR to heat stress., Pandey and Kalra (2005) examined that selected groups of resistance activator viz. Isonicotinamide, 2-chloronicotinic acid, 5-nitrosalicylic acid, 4-chlorosalicylic acid, DL-2 aminobutyric acid, 2-aminobutyric acid, O-acetylsalicylic acid, 4-amino salicylic acid, salicylic acid and these were used as soil drench using 3 week old seedlings transplanted to root-knot nematode infested pots. Maximum reduction in root-knot severity and nematode population occurred with 4-chlorosalicylic acid, O-acetyl salicylic acid, 2-chloronicotinic acid and gave significant flower yield advantages., Raju *et al.* (2008) revealed that the seeds of susceptible cultivar L-550 pretreated with salicylic acid (SA), spermine) Spm) and SA+Spm delay the onset of disease symptoms, whereas the seeds treated with water showed complete collapse within 25 days of treatment. During the present study also the application of

salicylic acid and isonicotinamide resulted in reduction of disease incidence. However the inhibitory effect on the pathogen was not observed. This supports the action of these two elicitors through imparting plant resistance rather than directly affecting the activity/the growth of the pathogen.

5.9 Influence of chemical elicitors on germination percent of chickpea in the poly house

The data presented in table 5 on the effect of 2 chemical elicitors in the chickpea for the control of dry root rot disease in poly house revealed non-significant differences in germination (%) by all the chemical elicitors over the control.

The data revealed that there were statistically non-significant differences between all the treatments but the highest germination was achieved by the seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm just after germination of the plants followed by seed treatment with 200ppm isonicotinamide and foliar application of isonicotinamide with 200ppm just after germination of the plants. There may not be the direct effect of elicitors on germination as they influence the physiology of the host for imparting resistance.

5.10 Height of chickpea plants as influenced by chemical elicitors

Height of plants (cm) was measured with the help of a meter scale. The data on this growth attributed character was recorded at harvesting time. The data showed that all the treatments showed non- significant variation in plant height.

The data revealed that there were statistically non-significant differences between all the treatments but the seed treatment with salicylic acid 200ppm resulted in maximum height of the plants.

5.11 Number of branches per plants as influenced by chemical elicitors

The data of number of branches per plot at the time of harvesting of chickpea is given in table 7. The number of branches enhanced tremendously with the increment in the growth and development of plant up to the harvesting.

Although there were statistically non-significant differences between all the treatments yet the application of salicylic acid 400ppm as seed treatment coupled with foliar application at 30 DAS resulted in maximum number of branches/plant. The chemical elicitors have earlier been reported to impart resistance to diseases although physiological branches in the plants when is also exhibited in their own physiological behavior.

5.12 Number of pods per plants as influenced by chemical elicitors

The number of pods per plant is an important yield attributing character ultimately influences the grain yield of any crop. The data on mean number of the pods per plant as affected by the different treatments is presented in table 8. The number of pods per plant was non-significantly affected by all the treatments.

Thus the application of salicylic acid 400ppm as seed treatment coupled with foliar application at 30 DAS resulted in maximum number of pods/plant.

5.13 100 grain weight as influenced by chemical elicitors

The data given in table 9 showed that only seed treatment with 400ppm salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS (14.31) influenced 100 seed weight of chickpea maximum but it was non-significant i.e. the treatments were at par with each other in respect of this character. However, the best treatment which enhanced the 100 grain weight was salicylic acid 400ppm coupled with its foliar application at 30DAS.

5.14 Yield of chickpea in the influence of chemical elicitors

The grain yield is the ultimate character for assessment of treatment in any agronomical investigation.

The data presented in table 4.10 revealed that all the treatments resulted in non-significant differences in the grain yield of chickpea. However, the maximum yield was obtained by the application of salicylic acid at 400ppm as seed treatment alone and coupled with its foliar application at

30DAS which was followed by seed treatment with isonicotinamide 200ppm coupled with foliar application 200ppm at 30DAS. With an overview of the observations on the development of the disease, growth the pathogen and plant characters and yield it can be inferred that even though the growth of the pathogen is not reduced by these elicitors yet the morphological attributes of the plants are observed to be better which ultimately impart slight resistance against the disease.

Chapter 6

Summary Conclusion and suggestions for Further work

The present studies on “chemical mediated induction of resistance in chickpea against dry root rot (*Rhizoctonia bataticola*).” through SAR/ISR compounds and response of these chemicals to different inoculum load and substratum were carried out in the laboratory, under pot conditions and in a field experiment. The results are being summarized as below.

6.1 Summary

- I. The typical symptoms of the disease and characterization of the pathogen were described to identify it as *R. bataticola*.
- II. Studies on control of dry root rot by seed treatments and foliar application of SAR compounds were undertaken. Among the tested treatments salicylic acid at 400ppm applied as seed treatment along with its foliar application at 30 DAS, isonicotinamide at 200ppm applied as seed treatment coupled with foliar application at 30 DAS in field condition and salicylic acid at 400ppm applied as seed treatment along with its foliar application just after germination of the plants, isonicotinamide at 200ppm applied as seed treatment coupled with foliar application just after germination of the plants in pot condition were most effective whereas seed treatments with 200ppm salicylic acid, seed treatments with 200ppm isonicotinamide were least effective treatment against *R. bataticola*.
- III. Experiments carried out to evaluate the resistance through salicylic acid and isonicotinamide reduced disease incidence caused by *R. bataticola* by imparting resistance to the host rather than directly affecting the pathogen.

6.2 Conclusion

Studies on effect of seed treatments and foliar application on the incidence of dry root rot in chickpea showed that the disease incidence was significantly reduced by combination treatment seed treatment with 400ppm

salicylic acid and foliar application of salicylic acid with 400ppm at 30 DAS, seed treatment with 400ppm isonicotinamide and foliar application of isonicotinamide with 400ppm at 30 DAS in field condition and salicylic acid at 400ppm applied as seed treatment along with its foliar application just after germination of the plants, isonicotinamide at 400ppm applied as seed treatment coupled with foliar application just after germination of the plants in pot condition were the best but salicylic acid at 400ppm applied as seed treatment along with its foliar application at 30 DAS was proved in best with respect to other control measures.

6.3 Suggestion

- I. The present findings on the basis of one year experimentation need further confirmation through field, pot and laboratory studies.
- II. Complementally information on wide chemical elicitors can be used to control of disease.
- III. Proper timing and appropriate combination of chemical elicitors should be exploited for effective management of seed and soil borne disease of chickpea.

6.4 Further work

The present investigation indicated that seed treatments and foliar application of chemical elicitors salicylic acid and isonicotinamide at different concentrations reduced disease incidence in chickpea in field experiment and pot culture which is a cheaper method. Efforts can also be made to explore still better and cheaper techniques to control the disease either through induction of resistance, VAM fungi etc. in future.

CHAPTER – 7

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Table: 1a Incidence of dry root rot in chickpea under the influence of chemical elicitors 30 DAS

Source of variance	DF	SS	MSS	F.cal
Replications	2	0.80	0.40	0.20
Treatments	8	153.59	19.20	9.73
Error	16	31.57	1.97	
Total	26	185.96		

Significant at 5% level

Table: 1b Incidence of dry root rot in chickpea under the influence of chemical elicitors 60 DAS

Source of variance	DF	SS	MSS	F.cal
Replications	2	8.59	4.29	1.72
Treatments	8	689.19	86.15	34.45
Error	16	40.01	2.50	
Total	26	737.79		

Significant at 5% level

Table: 1c Incidence of dry root rot in chickpea under the influence of chemical elicitors 90 DAS.

Source of variance	DF	SS	MSS	F cal
Replications	2	3.99	2.00	0.29
Treatments	8	1014.16	126.77	18.17
Error	16	111.61	6.98	
Total	26	1129.77		

Significant at 5% level

Table: 2 influence of chemical elicitors on germination percent of chickpea at 10 DAS.

Source of variance	DF	SS	MSS	F. cal
Replications	2	31.61	15.81	0.49
Treatments	8	440.70	55.09	1.70
Error	16	519.38	32.46	
Total	26	991.69		

Significant at 5% level

Table: 3 Radial growth of *R. bataticola* as influenced by chemical elicitors

Source of variance	DF	SS	MSS	F.cal
Treatments	8	93.18	11.65	28.63
Error	27	10.98	0.41	
Total	35	104.17		

Significant at 5% level

Table: 4a Incidence of dry root rot in chickpea under the influence of chemical elicitors 12 DAS in poly house:

Source of variance	DF	SS	MSS	F.cal
Treatments	8	270.52	33.82	3.34
Error	27	273.07	10.11	
Total	35	543.59		

Significant at 5% level

Table: 4b Incidence of dry root rot in chickpea under the influence of chemical elicitors 20 DAS in poly house:

Source of variance	DF	SS	MSS	F.ci
Treatments	8	965.61	120.70	12.32
Error	27	264.59	9.80	
Total	35	1230.20		

Significant at 5% level

Table: 4c Incidence of dry root rot in chickpea under the influence of chemical elicitors 27 DAS in poly house:

Source of variance	DF	SS	MSS	F.cal
Treatments	8	1335.99	167.00	16.58
Error	27	271.89	10.07	
Total	35	1607.88		

Significant at 5% level

Table: 5 Germination of chickpea seeds under the influence of chemical elicitors in the poly house 5 DAS:

Source of variance	DF	SS	MSS	F.cal
Treatments	8	754.34	94.29	7.56
Error	27	336.61	12.47	
Total	35	1090.94		

Significant at 5% level

Table: 6 Height of chickpea plants as influenced by chemical elicitors

Source of variance	DF	SS	MSS	F.cal
Replications	2	41.61	20.80	4.60
Treatments	8	30.37	3.80	0.84
Error	16	72.36	4.52	
Total	26	144.35		

Non-significant at 5% level

Table: 7 Number of branches per plants as influenced by chemical elicitors:

Source of variance	DF	SS	MSS	F.cal
Replications	2	2.47	1.23	5.08
Treatments	8	4.33	0.54	2.23
Error	16	3.88	0.24	
Total	26	10.68		

Non-significant at 5% level

Table: 8 Number of pods per plants as influenced by chemical elicitors

Source of variance	DF	SS	MSS	F.cal
Replications	2	197.66	98.83	3.96
Treatments	8	266.43	33.30	1.33
Error	16	399.54	24.97	
Total	26	863.63		

Non-significant at 5% level

Table: 9 100 grain weights as influenced by chemical elicitors.

Source of variance	DF	SS	MSS	F.cal
Replications	2	0.17	0.09	0.36
Treatments	8	2.71	0.34	1.39
Error	16	3.88	0.24	
Total	26	6.77		

Non-significant at 5% level

Table: 10 Yield of chickpea under in the influence of chemical elicitors (q/ha).

Source of variance	DF	SS	MSS	F.cal
Replications	2	11.38	5.69	0.91
Treatments	8	113.12	14.14	2.27
Error	16	99.74	6.23	
Total	26	224.24		

Non-significant at 5% level

VITA

The author of this thesis **Mr. Amrit Kumar Pipalde** S/o shri Bhadar pipalde born on 01th Apr 1987 at Khargone district in Madhya Pradesh. He completed his schooling from Mahatma Gandhi Govt. Higher Secondary School, at Mandleshwer in Khargone district, securing first division in Higher Secondary Certificate Examinations.

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of Masters Degree he was allotted a topic “**Studies on chemical mediated induction of resistance in chickpea against dry root rot (*Rhizoctonia bataticola*)**” which was successfully completed by him and presented in the form of this thesis.